Copernicus Global Land Operations
”Vegetation and Energy”
“CGLOPS-1”
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ALGORITHM THEORETHICAL BASIS DOCUMENT

DRY MATTER PRODUCTIVITY (DMP)
GROSS DRY MATTER PRODUCTIVITY (GDMP)
NET PRIMARY PRODUCTION (NPP)
GROSS PRIMARY PRODUCTION (GPP)
VERSION 2.0
Issue 1.00

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List of Acronyms

ACT	Actual
AD	Applicable Document
APAR	Absorbed Photosynthetically Active Radiation
AS	Age scalar
ATBD	Algorithmic Theoretical Basis Document
AVHRR	Advanced Very High-Resolution Radiometer
BEF	Broadleaved Evergreen Forests
BIOME-BGC	BioGeochemical Cycles model
BPLUT	Biome parameter look-up table
CASA	Carnegie-Ames-Stanford-Approach
CCI	Climate Change Initiative
C-fix	Carbon fixation model
Cflux	Carbon flux model
CL	Cloudiness
CLMS	Copernicus Land Monitoring Service
CGLOPS	Copernicus Global Land Operations
CSV	Comma Separated Value
CUE	Carbon Use Efficiency
DGVM	Dynamic Global Vegetation Model
DN	Digital Number
DM	Dry matter
DMP	Dry matter productivity
DTN	Data Transmission Network, United States private company
EC	Eddy Covariance
EC-LUE	Eddy covariance flux light use efficiency model
ECMWF	European Center for Medium range Weather Forecasting
ED	Ecosystem Demography
EF	Evaporative fraction
ESRL	Earth System Research Laboratory
ET	Evapotranspiration
fAPAR	Fraction of absorbed photosynthetically active radiation
FD	Frost days
GDMP	Gross Dry Matter Productivity
GLIMPSE	Global Image Processing Software
GLO-PEM2	global production efficiency model version 2

GPP	Gross primary production
GVMI	Global Vegetation Moisture Index
HRES	High Resolution Atmospheric Model of ECMWF
IGBP	International Geosphere-Biosphere Programme
ISLSCP	International Satellite Land-Surface Climatology Project Initiative
J_{AP}	Joule APAR
J_P	Joule PAR
Jt	Joule Total Radiation
JRC-MARS	Joint Research Centre - Monitoring Agricultural Resources
K	Kelvin
LAI	Leaf area index
LCCS	Land Cover Classification System
LPJ-DGVM	Lund-Potsdam-Jena Dynamic Global Vegetation Model
LUE	Light use efficiency
MODIS	Moderate Resolution Imaging Spectroradiometer
MSG	Meteosat Second Generation
NBP	Net biome productivity
NDVI	Normalized difference vegetation index
NEE	Net ecosystem exchange
NEP	Net ecosystem production
NIR	Near infrared
NRT	Near real time
NOAA-ESRL	NOAA Earth System Research Laboratory (ESRL)
NPP	Net primary production
OLCI	Ocean and Land Colour Instrument
ORCHIDEE	ORganizing Carbon and Hydrology In Dynamic EcosystEms
PAR	Photosynthetically Active Radiation
PEM	Production Efficiency Models
3-PGS	Physiological Principles Predicting Growth from Satellite
PET	potential evapotranspiration
PL	Leaf phenology
PRI	Photochemical Reflectance Index
PsnNet	Net photosynthesis
PUM	Product User Manual
R	Radiation
Ra	Autotrophic respiration

REGCROP	Regional Crop model
RMSE	Root Mean Squared Error
Rs	surface reflectance
SIF	Solar Induced Fluorescence
SLSTR	Sea and Land Surface Temperature Radiometer
SM	Soil moisture
SWIR	Short wave infrared
T_a	Air temperature
T_s	Surface temperature
VGT	SPOT-VEGETATION sensor
VITO	Vlaamse Instelling voor Technologisch Onderzoek
VLR	Very low resolution
VOD	Vegetation Optical Depth
VPD	Vapour Pressure Deficit
VPM	Vegetation Photosynthesis Model
W	Canopy water content

EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

Dry Matter Productivity (DMP) is one of the global bio-geophysical products delivered by the CLMS, representing the daily growth of standing biomass. From January 2014 to June 2020, the DMP 300m Version 1.0 product was derived using fAPAR Version 1.0 from PROBA-V sensor data. From July 2020 onwards, the production is continuing using fAPAR Version 1.1 derived from Sentinel-3/OLCI sensor data. It is calculated globally and made available to the user in near-real time (NRT) every 10 days. Next to DMP, the Gross Dry Matter Productivity (GDMP) is also provided. GDMP equals DMP without autotrophic respiration.

Net Primary Production (NPP) expresses the rate at which plants produce net useful chemical energy expressed in mass of carbon per unit area per day ($gC/m^2/day$). NPP is a rescaling of the DMP product to fulfill the needs of the climate community. Similar to GDMP, also GPP (Gross Primary Production) is provided, which is the NPP without considering the autotrophic respiration.

This document gives a theoretical basis of the DMP/GDMP and NPP/GPP version 2.0 retrieval, describes the algorithm, briefly assess its performance and its practical processing procedure.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

This document describes the theoretical background of the operational algorithm of the Dry Matter Productivity (DMP) / Gross Dry Matter Productivity (GDMP) version 2.0 and the Net Primary Production (NPP) / Gross Primary Production (GPP) version 2.0 as distributed in the Copernicus Land Monitoring Service (CLMS). In a more technical sense also the processing set up is outlined. Related input data are briefly discussed.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 is a review of the user requirements.
- Chapter 3 is an outline of the general context of ecosystem productivity and how it is calculated.
- Chapter 4 describes the DMP/GDMP and NPP/GPP product from the theoretical base to the operational set-up. The performance of the algorithm is briefly assessed.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_ATBD_FAPAR300m-V2.0	Algorithm Theoretical Basis Document of the fAPAR 300m, Version 2.0 derived from Sentinel-3 sensors data
GIOGL1_ATBD_DMP1km-V1	Algorithm Theoretical Basis Document of the DMP 1km, version 1
CGLOPS1_ATBD_DMP1km-V2	Algorithm Theoretical Basis Document of the DMP 1km, version 2

CGLOPS1_ATBD_FAPAR1km-V2	Algorithm Theoretical Basis Document of the fAPAR 1km, version 2
CGLOPS1_PUM_FAPAR300m-V2.0	Product User Manual of the fAPAR 300m version 2.0 derived from Sentinel-3 data

Documents available at https://land.copernicus.eu/en/technical-library

1.3.3 Output

Document ID	Descriptor
CGLOPS1_PUM_DMP-NPP300m-V2.0	Product User Manual summarizing all information about the DMP/GDMP and NPP/GPP 300m V2.0 products.
CGLOPS1_QAR_DMP-NPP300m_V2.0	Report presenting the results of the quality assessment of the DMP/GDMP and NPP/GPP 300m V2.0 product

Documents available at https://land.copernicus.eu/en/technical-library

1.3.4 External document

GCOS#245 The 2022 GCOS ECVs Requirements

Available at https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

2 USER REQUIREMENTS

The user requirements relevant for DMP and NPP are according to the applicable document [AD1].

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Baseline dataset pixel resolutions	300m
Target baseline location accuracy	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetical datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic Coverage:	Global
Temporal resolution	10-day period (dekad: days 1-10, 11-20, 21 end of month)
Timeliness	Within 2 days (optimally 1 day) after the end of each dekad

DMP – DRY MATTER PRODUCTIVITY	
Amount (weight) of dry matter (DM) produced per surface unit and per time unit expressed in kilograms of dry matter per hectare per day (kgDM/ha/day). The accumulation over time leads to the assessment of the biomass (in this context, phytomass), and can be defined as above-ground mass per unit area of living plant material (GCOS 245).	
Uncertainty (2-sigma)	Threshold: 20% (can be biome specific: <10% in crop areas; 20% for grassland); Goal: 10% (can be biome specific: <10% in crop areas; 20% for grassland) [kg DM/ha/day]
Stability (per decade, i.e. 10 years)	Threshold: 10% Goal: 5%

NPP – ABOVE GROUND NET PRIMARY PRODUCTION	
Net primary production is the rate at which plants produce net useful chemical energy expressed in mass of carbon per unit area per day (gC/m²/day)	
Uncertainty (2-sigma)	Threshold: 20% (can be biome specific: <10% in crop areas; 20% for grassland);

	Goal: 10% (can be biome specific: <10% in crop areas; 20% for grassland);
Stability (per decade, i.e. 10 years)	Threshold: 10% Goal: 5%

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets shall be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.

The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 GENERAL CONTEXT – ECOSYSTEM PRODUCTIVITY

3.1 INTRODUCTION

The productivity of an ecosystem is a fundamental ecological variable, which measures the energy input to the biosphere and terrestrial carbon dioxide assimilation. Monitoring ecosystem carbon fluxes is key in mitigating global change and widely used in climatology studies. In general, ecosystem productivity is a good indicator for the condition of the land surface area and status of a wide range of ecological processes (Gower et al., 2001). Ecosystem productivity is also a unique integrator of climatic, ecological, geochemical and human influences (Running et al., 2009) and a suitable indicator for agronomic analyses.

Ecosystem productivity is a complex and dynamic system characteristic influenced by many factors. Theoretically it can be expressed in the form of carbon fluxes. Figure 1 gives a schematic representation of the different component fluxes and processes. *Gross Primary Production* denotes the total amount of carbon fixed in the process of photosynthesis by plants in an ecosystem. A large fraction (up to half of it) is immediately lost again through the process of autotrophic respiration required to provide the necessary energy for maintenance and growth of biomass. The remainder is called *Net Primary Production.* NPP can be considered as the short-term carbon storage. On a slightly longer term, there is decomposition (heterotrophic respiration) of litter and soils in excess of that resulting from disturbance further releases CO2 to the atmosphere, reducing NPP to *Net Ecosystem Production* and resulting in medium-term carbon uptake. Disturbance from both natural (e.g. fire, flooding, landslides) and anthropogenic sources (e.g., harvest) leads to further release of CO2 to the atmosphere by additional heterotrophic respiration and combustion-which, in turn, leads to long-term carbon storage (*Net Biome Production*). Satellite based estimates of primary production focus on GPP and NPP. The global Dry Matter Productivity (DMP) of the Copernicus Land Monitoring Service is closely related to GPP and NPP (see side box below for a further clarification).

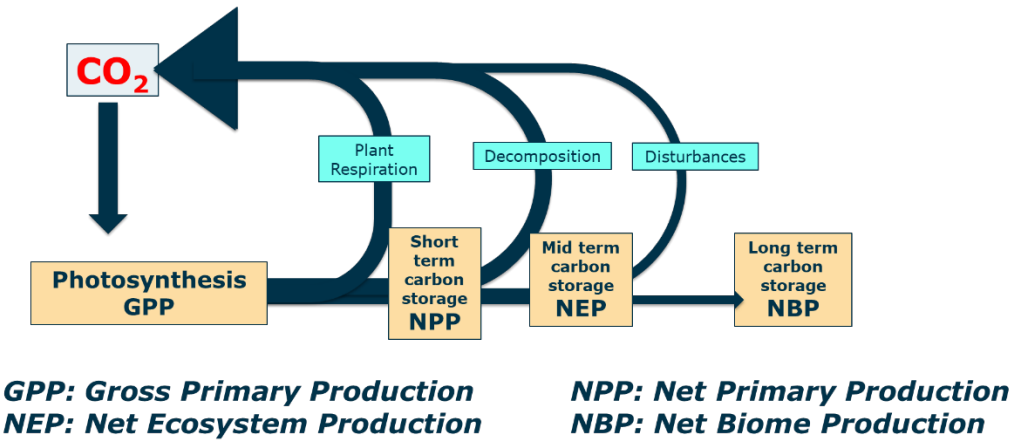


Figure 1: The component fluxes and processes in ecosystem productivity. GPP: Gross Primary Production, NPP: Net Primary Production, NEP: Net Ecosystem Production, NBP: Net Biome Production (Valentini, 2013)

Side Box: Dry Matter Productivity (DMP) explained

Defining Dry Matter Productivity (DMP)

Dry Matter Productivity (DMP) and Gross Dry Matter Productivity (GDMP) are two components expressing components of Primary Production of ecosystems related to the creation of new organic matter by vegetation. GDMP is (similar as GPP) the total amount of dry matter "fixed" by land plants per unit time through photosynthesis (atmospheric CO2 into organic compounds). A substantial fraction of GDMP supports plant autotrophic respiration (Ra), with the remainder allocated to the Dry Matter Productivity (DMP, similar as NPP) of plant structural biomass in stems, leaves, and fruit, and, to a much lesser extent, volatile organic compounds used in plant defense and signaling (Gough, et al., 2011). GDMP and DMP both represent the above-and belowground part of plants. To convert DMP into the aboveground part a root to shoot ratio should be applied on the DMP values.

Gough, C. M. (2011) Terrestrial Primary Production: Fuel for Life. Nature Education Knowledge 3(10):28

Relation between Dry Matter Productivity (DMP) and Gross (GPP)/Net (NPP) primary production

The DMP product contains two layers named DMP (Dry Matter Productivity) and Gross Dry Matter Productivity (GDMP). These are in fact equivalent of respectively NPP (for DMP) and GPP (for GDMP). The only difference (expect for the naming) is that DMP is expressed in kg Dry Matter per hectare per day (kg DM/ha/day) while GPP/NPP are typically expressed as gram carbon per square meter per day (C/m2/day). The name and unit choice of DMP was originally designed to align with the terminology used in agricultural monitoring applications. When the DMP algorithm and product were adopted by the Copernicus Land Monitoring Service, the name and unit where adopted. But, in fact, DMP could be better called Dry Matter Production (instead of Productivity). To convert the values and units of DMP to NPP or GDMP to GPP a simple conversion factor can be applied:

First, kg DM/ha/day can be converted into g DM/m2/day since:

1 kgDM/ha/day = 1000 gDM/ha/day = 0.1 gDM/m^2/day

For the conversion of Dry Matter (DM) into Carbon (C) the conversion factor defined in the IPCC Good Practice Guide (IPCC, 2003) can be used which states that vegetation dry matter generally consists of 50% of carbon. So gC/gDM = 0.5. So, to summarize, NPP and GPP are calculated from DMP and GDMP as follows:

NPP [gC/m^2/day] = DMP [kgDM/ha/day] × 0.5 × 0.1

GPP [gC/m^2/day] = GDMP [kgDM/ha/day] × 0.5 x 0.1

3.2 ALTERNATIVE METHODOLOGIES IN USE

Different methodologies exist for estimating Ecosystem Productivity, e.g. inventory analysis, micrometeorological measurements, empirical models, biogeochemical models, dynamic global vegetation models, and remote sensing (Ito, 2011).

3.2.1 Inventory-based methods

On-the-ground inventory-based methods are used in cropland, grassland and forested ecosystems. Ecosystem productivity can be estimated through periodic measurements of root, stem, leaf and fruit. In forested biomes, allometric functions are used to estimate the total tree biomass. These inventory based models prove their usefulness for sites studies at small scales and are capable to include anthropogenic disturbances, i.e. estimates up to the level of Net Biome Production (Nabuurs et al., 2003). However, they lack power for generalization and upscaling and are dependent on the quality and frequency of the inventories.

3.2.2 Eddy covariance measurements

FLUXNET (https://fluxnet.org) is a global network of micrometeorological flux measurement sites that use eddy covariance techniques to measurements of ecosystem variables like water, carbon, energy and nutrient fluxes. Working on a continuous basis, these towers provide data at a high temporal frequency (typically 30 min). Meteorological towers measure net ecosystem CO_2 exchange (NEE), which is equal to GPP minus ecosystem respiration and inorganic fluxes of CO_2. GPP estimates are derived from the NEE measurements by means of an ecosystem respiration model. The difficulty is to extrapolate from these footprint values to regional or global scales. Artificial intelligence and neural networks are gaining importance in this matter. The largest and most successful implementation of machine learning techniques for primary production modelling is the FLUXCOM initiative. Their approach makes use of the global FLUXNET in-situ data of GPP and NEE (Net Ecosystem Exchange) as reference and target variables. Different machine learning methodologies such as tree-based methods, regression splines, neural networks and kernel methods are to build the fitting models. Two setups are supported. The first delivers output GPP and NEE images at 0.0833°resolution using MODIS remote sensing data and the second delivers 0.5° resolution using remote sensing and meteorological data. A limitation of machine learning approaches is that they often do not capture the long terms trends such as the increase in primary productivity due to the atmospheric CO2 fertilization effect (Yang, et al., 2022). Biome-specific Light Use Efficiency (LUE) values can also be derived indirectly from flux tower eddy covariance (EC - indirect) measurements. The EC method determines carbon fluxes through the covariance between fluctuations in vertical wind profile and the CO_2 mixing ratio in the air above the canopy.

3.2.3 Inverse modelling

The inverse modelling approach ('top down') uses the spatial and temporal measurements of atmospheric CO_2 in conjunction with atmospheric transport modelling to estimate net carbon fluxes of ecosystems (Ciais et al., 2000; Rivier et al., 2010). A problem with this approach is the coarse resolution of the measurement network of atmospheric CO_2 and the possibility to separate anthropogenic and biogenic fluxes (Miglietta et al., 2007).

3.2.4 Dynamic global vegetation models (DGVM)

DGVMs are models that simulate ecosystem processes as a response to climate. Biogeochemical and hydrological cycles are modelled giving the constraints of latitude, topography and soil characteristics. Time series of climate data feed the models to produce gridded output at various temporal (hours to months) and spatial (30 arc-seconds to several degrees) scales. Several DGVMs have been developed by various research groups around the world that produces GPP and NPP estimates, e.g. ORCHIDEE (Krinner et al., 2005), JULES (Clark et al., 2011), LPJ-DGVM (Sitch et al., 2003), etc.

3.2.5 Satellite based Production Efficiency Models (PEM)

Production efficiency models (PEMs) are based on the theory of light use efficiency (LUE) which states that a relatively constant relationship exists between photosynthetic carbon uptake and radiation receipt at the canopy level (McCallum et al., 2009). Solar radiation is the main driver, downscaled by a number of efficiency factors being satellite based (fraction of Absorbed Photosynthetically Active Radiation, fAPAR) or modelled via environmental inputs such as temperature, soil moisture, etc. The CLMS global DMP, described in this document, is a product of a typical PEM model. Other PEM models are GLOPEM, C-FIX and the MODIS PEM model. A more detailed review is given in section 3.3.

3.2.6 Earth Observation (EO) proxies for photosynthesis

SIF (Solar Induced Fluorescence) has emerged as an important new remote sensing tool to study terrestrial vegetation (Mohammed et al. 2019; Porcar-Castell et al. 2014) and is increasingly used to study large scale photosynthesis (Ryu, et al., 2019). SIF has the unique feature of being a signal that is emitted from plant leaves, rather than only reflected, which has two main implications for its relevance for remotely monitoring plant photosynthesis. First, the signal is specific to vegetation, i.e., it is not affected by soil or other background signals. Second, the emission of SIF is mechanistically linked to photosynthesis. The relationship between SIF and photosynthetic LUE is non-linear at the leaf-level but tends to linearize at the canopy level and longer time scales (Zhang et al. 2016). Therefore, SIF has a strong potential to be used for photosynthesis monitoring at large scales. While SIF has been proposed for large scale GPP modelling, there are still open research questions on the

SIF to GPP relation such as the portioning into structural and physiological component (Dechant et al. 2020) and the response to vegetation stresses (Wohlfahrt et al. 2018, Parazoo et al. 2020).

Vegetation Optical Depth (VOD) is a key parameter in describing vegetation attenuation properties in microwave remote sensing. It is considered as a proxy of vegetation properties (which are typically not captured by traditional vegetation indices) such as tree height, sap flow, leaf fall, aboveground biomass and also GPP. Since it is derived from passive and active microwave observations it is not affected by the presence of clouds. Recently, efforts have been made to use VOD to generate global GPP datasets (Wild, et al. 2022).

3.2.7 DMP relation to other sources

The CLMS global DMP is a Monteith variant (see 3.3) of the Production Efficiency Models (PEM). Its output is most comparable against other satellite-based PEM such as the MODIS GPP/NPP. The DGVMs are also produced at large scale (regional to global) but the time step is denser and the modelling is often more complex, e.g. by including productivity constraints such as water and nutrient deficiency, climate feedback cycles, etc. Their output cannot be compared directly with the data driven PEM approach, which estimates potential rather than actual vegetation production. The same holds for the inverse models, which are more suited for climate studies than for operational vegetation monitoring. The inventory-based models are also not directly related to the DMP, as their focus is to deal with detailed studies which require extensive local inputs. The eddy covariance measures, such as installed on the FLUXNET towers, estimate NEE but can be converted to GPP (or GDMP). Despite their different spatial footprint and other constraints, they are considered as valuable input for any satellite based GPP/NPP models and are often used for the calibration and validation of such models.

3.3 MONTEITH APPROACH

3.3.1 Outline

The global DMP product of the Copernicus Land Monitoring Service is based on the Light Use Efficiency (LUE) approach first formulated by Monteith (1972). He stated that the vegetation growth is completely defined by the part of the incoming solar radiance that is used for photosynthesis and which is absorbed by the plants (Absorbed Photosynthetically Active Radiation, APAR, [$kJ_{AP}/m^2/d$]) and an actual conversion efficiency factor ε_{ACT}.

$$\boldsymbol{NPP = APAR \times \varepsilon_{ACT}} \qquad \textbf{(1)}$$

The fraction of absorbed photosynthetically radiation (fAPAR, [J_{AP}/J_P]) is used in combination with the incident solar radiation from meteorological data (R, [$kJ/m^2/d$]) to form APAR in equation (1).

fAPAR can be estimated from the reflectance information derived from optical sensors that have at least spectral bands in the red and near infrared part of the solar spectrum (Baret et al., 2007; Weiss et al., 2010). In the previous equation, the term APAR is replaced by:

$$\boldsymbol{APAR = R \times fAPAR \times \varepsilon_c} \quad \textbf{(2)}$$

The term ε_c describes the portion of the intercepted solar radiation that is potentially suitable for photosynthesis, i.e. the climatic efficiency. In models, this is often a constant fraction, whereas in reality it depends on location and sky conditions (Gower et al., 1999).

The energy conversion factor ε_{ACT} in equation (1) expresses the actual efficiency of converting atmospheric CO_2 into plant tissue. This actual efficiency can be subdivided into a vegetation type specific maximum light use efficiency terms ε_{LUE} and a number of stress factors. The term ε_{LUE} is then the light-use efficiency in optimal conditions, i.e. when there is sufficient water and nutrients, the temperature is optimal for vegetation growth, no pests, diseases, etc. In the DMP, this ε_{LUE} is biome specific. All these limiting factors can be used to downscale ε_{LUE} towards ε_{ACT}.

Gross Primary Productivity is obtained when the respiration processes are not taken into account. The equation is as follows:

$$\boldsymbol{GPP = R \times fAPAR \times \varepsilon_c \times \varepsilon_{LUE} \times \varepsilon_T \times \varepsilon_{H2O} \times \varepsilon_{CO2} \times \varepsilon_{res}} \quad \textbf{(3)}$$

The different conversion and efficiency terms are explained in Table 1. The final equation (1) then becomes:

$$\boldsymbol{NPP = R \times fAPAR \times \varepsilon_c \times \varepsilon_{LUE} \times \varepsilon_T \times \varepsilon_{H2O} \times \varepsilon_{CO2} \times \varepsilon_{CUE} \times \varepsilon_{res}} \quad \textbf{(4)}$$

Table 1: Definition of the different conversion and efficiency terms in equation (4).

TERM	*MEANING*	*VALUE*	*UNIT*
ε_c	Climatic efficiency, i.e. the fraction of photosynthesis active radiation (PAR) in total shortwave radiation R. This depends on location and on sky conditions (Gower et al., 1999). In most models only one invariant value is used for ε_c.	0.42 – 0.55	J_P/J_T
ε_{LUE}	Maximum light use efficiency, i.e. in optimal conditions with no limitations for vegetation growth. It expresses the efficiency with which vegetation performs photosynthesis.	Biome-specific LUE	g C/kJ_{AP}
ε_T	Efficiency term limiting the vegetation growth for temperature stress. The term is normalized between 0 and 1.	0 – 1	-
ε_{H2O}	Efficiency term limiting the vegetation growth for water stress. The term is normalized between 0 and 1.	0 – 1	-
ε_{CO2}	CO_2 fertilization effect, normalized between 0 and 1	0 – 1	-
ε_{CUE}	Carbon Use Efficiency (CUE)	0 – 1	-
ε_{res}	Fraction kept after residual effects (pests, lack of nutrients, etc.)	0 – 1	-

3.3.2 Basic underlying assumptions

The Monteith approach assumes radiation, temperature and water availability are the fundamental physiological limiting factors for NPP. Figure 2 illustrates the spatial distribution of these limits on a global scale. Note that residual stress factors like water stress, diseases or nutrient deficiencies (ε_{res}) are not directly accounted for in the CLMS DMP algorithm. Such residual stress factors are, however, intrinsically present in the fAPAR data.

The accuracy of GPP estimates depends on the accuracy with which the compounding factors can be measured or estimated using meteorological, ecophysiological and remotely sensed data. Much of the uncertainty in estimating GPP is due to variability in estimates of solar radiation conversion to biomass, the light use efficiency factor. This efficiency term is critical for GPP estimation, yet the controls on ε_{LUE} are still poorly understood (Jenkins et al., 2007). Uncertainties are associated with factors such as canopy chemistry and structure, respiration costs for maintenance and growth, canopy temperature, evaporative demand, and soil water availability (Running et al., 2009). Another question on the downregulation of ε_{LUE} is whether it should be downregulated by all stressors or by the most limiting ones. A challenge at global scale is assessment of the maximum ε_{LUE} (i.e. in optimal conditions) per land cover type or plant functional type. Some authors even question whether this is necessary (e.g. Yuan et al., 2007).

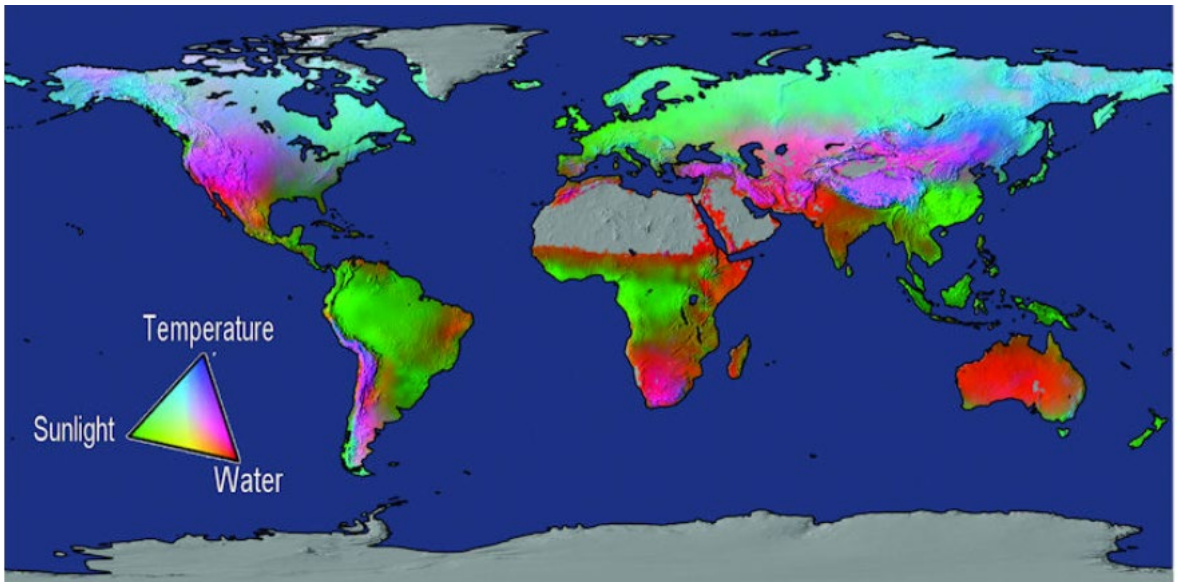


Figure 2: Potential limits to vegetation net primary production based on fundamental physiological limits by solar radiation, water balance, and temperature (from: Churkina and Running, 1998; Nemani et al., 2003; Running et al., 2004). Retrieved from http://www.ntsg.umt.edu/project/mod17.

3.3.3 Controls of Light Use Efficiency (LUE)

During the past decade, many researchers have investigated the controls of the light use efficiency term ε_{LUE} (e.g., Garbulsky et al., 2010; Turner et al., 2003; Wang et al., 2010). This is important in two respects: first to derive the maximum gross light efficiency factor ε_{LUE}, and secondly to know which parameters have to be used in order to downregulate ε_{LUE} to ε_{ACT} (the actual efficiency) in GPP-models. So far, the controls on ε_{LUE} are still poorly understood, which is reflected in the multitude of different GPP-models that exist (see next section).

Garbulsky et al. (2010) performed an analysis based on data from 35 eddy covariance (EC) flux sites comprising in total 90 site/years of data, from a wide range of climatic conditions. ε_{LUE} was calculated from EC measurements and MODIS fAPAR data and inter-annual and intra-annual controls on ε_{LUE} were investigated. They concluded that when vegetation is adapted to its local environment, water availability is more constraining its functioning than temperature. This condition is in agreement with the results of Yuan et al. (2007), who found that ε_{ACT} is predominantly controlled by moisture conditions throughout the growing season and that the temperature is only important at the beginning and the end of the growing season.

To explain the spatial variability of ε_{LUE}, Garbulsky et al. (2010) found that annual precipitation is more important than vegetation type. Although not the most important factor, they did conclude that

vegetation type matters, which is in line with the former results of Turner et al. (2003), who support the idea of biome-specific parameterization of ε_{LUE}. The results of Wang et al. (2010) and Ahl et al. (2004) suggest that biome-specific ε_{LUE} are even not detailed enough, because of the heterogeneity of these classes. They argue for species-specific ε_{LUE}. In strong contrast to these findings are the results of Yuan et al. (2007), who claim that their model, using a single biome-independent ε_{LUE} outperforms the MODIS GPP data set using biome-specific ε_{LUE} values, and this based on validation data of 28 eddy covariance flux towers at diverse locations.

When looking at intra-annual variability of ε_{LUE}, Garbulsky et al. (2010) found a stronger link with energy-balance parameters (evaporative fraction) and water availability along a climatic gradient, but only a weak influence from vapour pressure deficit and temperature.

The main control parameters of the temporal evolution of ε_{LUE} varied across ecosystems. One example is that temperature only played a role in the intra-annual variability of ε_{LUE} at the coldest and energy-limited sites.

For humid and hot ecosystems, Garbulsky et al. (2010) concluded that none of the variables analysed were robust surrogates for ε_{ACT}. They suggest that the ratio of direct to diffuse light might play a more important role in the control of ε_{LUE}. Jenkins et al. (2007) has previously concluded from the time series analysis of measurements of one eddy covariance flux site that the day-to-day variation in ε_{LUE} is largely explained by changes in the ratio of diffuse to total downwelling radiation. They found no strong correlation with any other measured meteorological variable. Turner et al. (2003) found higher ε_{LUE} values in overcast conditions, and therefore suggest the inclusion of parameters on cloudiness in GPP-models.

Other results from Turner et al. (2003) suggest that the phenological status of the vegetation should be used as a parameter for GPP-estimation, because ε_{LUE} declined toward the end of the growing season for the agricultural site, which was attributed to a decrease in foliar nitrogen concentration. Also the development stage or stand age of forests is assumed to play an important role in the uptake of carbon (Law et al., 2001).

DeLucia et al. (2002) investigated the effect of elevated CO_2 concentrations on ε_{LUE} for forest plots using a free-air CO_2 enrichment system. They observed an only slight effect on leaf area index and no effect on the patterns of aboveground biomass allocation.

Finally, Gower et al. (1999) state that controls on ε_{LUE} should be expressed in function of GPP and not NPP, since the dependence of respiration processes on temperature and other controlling factors is different.

3.3.4 How is the LUE used in other models?

At present, a number of NPP models (Table 2) exist that make use of coarse spatial resolution optical remote sensing data. These models differ in many respects although they are all based on the general approach defined by Monteith (1972). Most of the current generation of light use efficiency models has 3 key components (Running et al., 2009):

(1) Satellite derived vegetation properties: land cover, LAI, fAPAR
(2) Daily climatic data including incident radiation, air temperature, humidity and rainfall
(3) A biome-specific parameterization scheme to convert absorbed PAR to NPP (ε_{ACT}).

Table 2: Examples of remote sensing driven GPP/NPP-models

Model	*Stressors*	ε_{LUE}	*Reference*
GPP-models			
GLO-PEM2	T_s, SM, VPD		(Goetz et al., 2000)
MODIS-PsN	T_s, VPD	Biome defined maximum (11 biomes)	(Zhao et al., 2005)
3-PGS	Soil, T_s, FD, P, WC	Function of soil nitrogen content	(Coops et al., 2010)
VPM	T_s, W, P_L	Fixed value, estimated from eddy covariance measurements	(Xiao et al., 2005)
EC-LUE	T_a, EF	Fixed value, estimated from eddy covariance measurements	(Yuan et al., 2007)
C-fix	T_s, EF, SM	Biome defined minimum/maximum	(Verstraeten et al., 2006)
CFlux	Ts, VPD, SM, CL, AS	Biome defined maximum in clear sky conditions	(King et al., 2011)
Copernicus Land Monitoring service	T_s	Biome defined maximum	(Veroustraete et al., 2002)
NPP-model			
CASA	T_s, SM T_s, ET, PET		(Potter et al., 1993)

PARAMETERS

AS	Age scalar
T_s	surface temperature
T_a	air temperature
SM	soil moisture
C	Cloudiness
VPD	vapour pressure deficit
P_L	Leaf phenology
EF	evaporative fraction
FD	frost days
W	canopy water content
ET	evapotranspiration
PET	potential evapotranspiration

MODELS

GLO-PEM2	global production efficiency model version 2
MODIS-PsN	MODIS - photosynthesis product
3-PGS	Physiological Principles Predicting Growth from Satellite
VPM	Vegetation Photosynthesis Model
EC-LUE	eddy covariance flux light use efficiency model
C-fix	Carbon fixation model
Cflux	Carbon Flux model
CASA	Carnegie-Ames-Stanford-Approach (different versions of this model exist)

Here we focus on the difference in the way ε_{LUE} is downregulated to ε_{ACT} using estimators of different stressors. The conversion efficiency term ε_{LUE} is usually expressed in terms of GPP and not NPP, because this allows a better incorporation of respiration processes in the model. Therefore, the models are subdivided in these two classes.

A number of studies compared the performance of different GPP/NPP models with and without input from remote sensing (e.g., Coops et al., 2009; Cramer et al., 1995; Ruimy et al., 1999; Tao et al., 2005; Wu et al., 2010). These models can differ in many respects. First, not all models downregulate the ε_{LUE} for the same stressors. The simplest model is only temperature based and only a few models

incorporate data on nutrient availability or on frost days. Most recent models include a water balance. Second, some models estimate the same stressors in different ways. The best example is the water limitation, which is realized through water vapour pressure deficit, evaporative fraction, soil moisture content, water holding capacity, evapotranspiration, or a combination of a set of these parameters. Third, input for the same parameter can be acquired via meteorological data or via satellite retrieval, e.g. temperature, vapour pressure deficit, soil water capacity.

When a comparison is made between two versions of a model, i.e. one without remote sensing data and one with (e.g. 3-PG vs 3-PGS, Sim-CYCLE vs MOD-Sim-CYCLE), then the version that uses remote sensing data performs better when compared with validation data (Hazarika et al., 2005).

An important issue that is not related to the efficiency term concerns the higher quality of retrievals from MODIS or SPOT-VGT data compared with AVHRR (e.g., Chiesi et al., 2005; Maselli et al., 2006).

Coops et al. (2009) compared the output of 3-PGS, MODIS and C-fix (a former version of the methodology described in this ATBD) for the United States and found that the differences among the models vary up to 50% in areas where topography and climate are more extreme. For the other areas, the variation was confined to 10%.

3.3.4.1 Estimation of maximum LUE

The light use efficiency (LUE) term ε_{ACT} is a value that represents the actual efficiency of a plant's use of absorbed radiation energy to produce biomass. It is therefore a key physiological parameter at the canopy scale. Production-efficiency models (such as the Monteith model) are based on the principle of downregulating a potential LUE with different stressors. The debate on whether this LUE should be a global constant (e.g. Veroustraete et al., 2002; Yuan et al., 2007), defined per biome (e.g., Garbulsky et al., 2010; Turner et al., 2003) or even species specific (Ahl et al., 2004; Wang et al., 2010) is still ongoing. But many of the current existing GPP models favour the biome-specific parameterization as a practical compromise between including mechanistic realism and avoiding too many parameters and detail for global scale applications. Different approaches exist to estimate Light Use Efficiencies; a number of these methods are described below.

Maximum gross LUE is commonly estimated indirectly using Eddy Covariance (EC) flux tower measurements, either by interpreting the tower measurements or by using the towers' GPP estimates to optimize the LUEs of existing GPP models. Figure 3 shows an overview of maximum LUE values estimated by different indirect approaches which are described below.

The term ε_{LUE} can then be derived indirectly using the Monteith approach, i.e. ε_{LUE} = GPP/APAR. This determination should be done over a certain period of time, from which a general ε_{LUE} can be

derived. Garbulsky et al. (2010) is an example of this approach by using the towers GPP data and MODIS fAPAR data to estimate biome-specific LUE values.

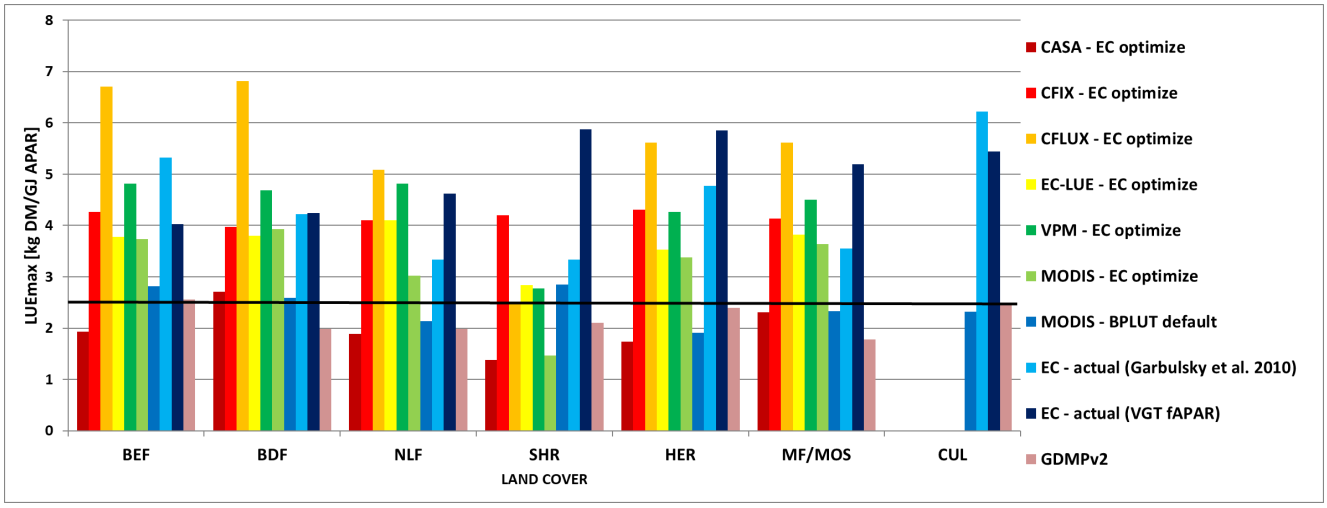


Figure 3: Estimated biome-specific gross maximum Light Use Efficiencies (LUE_{max}) with different approaches. The 'EC optimize' are model specific calibrated values against Eddy Covariance flux tower measurements. The values for the models CASA, CFIX, CFLUX, EC-LUE, VPM and MODIS are derived from Yuan et al. (2014). The MODIS – BPLUT are the default values used in the MOD17A2 product, adopted from Heinsch et al. (2003). The 'EC – actual' values are retrieved from flux tower measurements by Garbulsky et al. (2010) and own calculations using the SPOT-VGT fAPAR. The GDMPv2 are the values derived from a calibration of the 1 km GDMPv2 [CGLOPS1_ATBD_DMP1km-V2] with FLUXNET data. The current 300 m DMP adopted these values. The black horizontal line indicates the global constant 2.54 kg DM/GJ APAR of the DMP 1km version 1 [GIOGL1_ATBD_DMP1km-V1]. The biomes are BEF: Broadleaf Evergreen Forest, BDF: Broadleaf Deciduous Forest, NLF: Needleleaf Forest, SHR: Shrubland, HER: Herbaceous, MF/MOS: Mixed Forest/Mosaic Landscape, CUL: Cultivated.

Although widely used, methods using flux tower information suffer from a number of limitations and uncertainties. First, the flux footprint, which is captured in the eddy covariance sample area, is restricted to a few 100m² upwind from the tower. Consequently, the size and location of the footprint varies in time, making it less suitable to measure ε_{LUE} at a single canopy or small forest level. In addition, the eddy covariance theory assumes steady environmental conditions and surface homogeneity, a condition which is often violated.

Second, through the eddy covariance method, Net Ecosystem exchange (NEE) is measured, and not the desired GPP to derive ε_{LUE}. But for terrestrial ecosystems, one can assume that inorganic sinks and sources can be neglected (Lovett et al., 2006), meaning that –NEE equals Net Ecosystem Productivity (NEP). Then to obtain GPP, the NEP is summed with the daytime ecosystem respiration (R_d). This quantity however, should be measured in the absence of photosynthesis, so night time measurements are used, despite studies demonstrating the problems of extrapolation of night time measurements to daytime (Coops et al., 2010).

Third, radiation intercepted by non-photosynthetic plants are not part of NPP (by definition), but dead leaves do absorb PAR. As a result, one can observe an apparent reduction of ε_{LUE} near the end of the growing season by using the Monteith approach to derive ε_{LUE} from eddy covariance measurements (Gower et al., 1999). Thus, the period from which ε_{LUE} is derived is also important. At last, it is clear that the eddy covariance method provides actual light use efficiency terms and not the maximum gross light use efficiency term needed in NPP-models based on remote sensing. However, if the measurement period is sufficiently long, one could derive the gross light use efficiency, by analyzing all limiting factors within the same time span. Garbulsky et al. (2010) solved this problem by using the maximum occurring LUE value, assuming that at least once in the time series of flux data, the optimal conditions for vegetation growth were achieved. But caution should be taken as flux measurements often contain outliers.

Also, the global LUE constant of 2.54 $kgDM/GJ_{AP}$, used in the first version of the DMP originates from flux data with the above-described method. This value was adopted from Wofsy (1993) who calculated a maximum LUE based on April 1999 – December 1991 Eddy Covariance (EC) data (NEE, estimated Respiration and PAR) from a flux site at Harvard Forests.

A common approach to estimate biome-specific LUEs in GPP models is calibrating the model's output against flux tower GPP measurements hereby optimizing the LUEs per biome. Yuan et al. (2014) calibrated biome-specific LUE values of seven GPP models (CASA, Cfix, Cflux, EC-LUE, MODIS, VPM) using 157 eddy covariance (EC) towers by cross-validation, each time with 50% of the towers for calibration and 50% for validation. The advantage of such approach is that the estimated LUEs are tuned specifically for the respective model. The drawback however is that the optimized LUE values are dependent on the source of the input data.

The MODIS GPP (MOD17A2) algorithm also makes use of optimized maximum LUE values per biome type. The parameters in the Biome Property Lookup Table (BPLUT) are determined with BIOME-BGC simulations in conjunction with the validation data derived from GPP measurements from the BigFoot project. Chen et al. (2014) investigated how well an updated parametrization of these BPLUT parameters could improve the MODIS GPP estimations. Using Eddy Covariance data, they obtained more accurate GPP measurements with optimized LUE values compared to the BPLUT defaults.

Gross maximum LUE values estimated by the above methods are plotted per biome in Figure 3. In general, significant differences are shown between the different methods and models. The CASA model shows the LUEs representative for GPP, hence they cover a different order of magnitude. It is also clear that estimating the maximum LUE from time series of flux tower data ("EC – actual") yields higher values than the values used in the current GDMPv2. Furthermore, the optimized values per biome can differ significantly per model. Logically, as each model is built with its own stressors to downscale the maximum LUE (see 3.3.4). This line of reasoning implies that it is not feasible to define universal standard biome-specific maximum LUEs that can be adopted by any GPP model. Hence, it was decided in the CLMS DMP version 2 at 1 km (CGLOPS1_ATBD_DMP1km-V2) to derive optimized values, tuned specifically for DMP algorithm. Further details of this parameterization are discussed in "4.3 Methodology". These values are adopted by the 300 DMP.

Alternative approaches of estimating Light Use Efficiency (LUE) are:

- Photochemical Reflectance Index (PRI): PRI is a spectral index has been used widely as a proxy for LUE. PRI performed well in tracking LUE from leaf to canopy to ecosystem scale (Coops et al., 2010; Grace et al., 2007; Liu and Cheng, 2010; Garbulsky et al., 2014). Despite the promising approach, the upscaling to ecosystem level is still a challenge (Vanikiotis, et al., 2021).
- LUE can be dynamically modelled with first order ecological principles. The P-model for example combines the Farquhar-von Caemmerer-Berry model for C3 photosynthesis with an optimality principle for the carbon assimilation-transpiration trade-off and, predicts a multi-day average light use efficiency (LUE) for any climate and C3 vegetation type. (Stocker et al., 2020).

3.3.4.2 Water limitation

Most recent models include an explicit water limitation factor to incorporate drought effects in regulating the maximum LUE. According to Churkina et al. (1999), the approaches to introduce water budget limitation in NPP models can by divided in three modelling groups:

- Canopy conductance control on evapotranspiration
- Climatological supply/demand control on ecosystem productivity
- Water limitation inferred from satellite data

Canopy conductance is the upscaling of leaf stomatal response to water availability. In well-hydrated leaves, stomata are open, allowing CO_2 uptake and water vapour exiting the leaf. Decreasing the water content of leaves forces the stomata to close and limits photosynthesis. This behavior can be described as a function of incident radiation, vapour pressure deficit (VPD), air temperature, leaf

water potential and leaf area index. In the MOD17A2 algorithm, the VPD is used to include water stress in the GPP algorithm. The MODIS VPD is derived from daily meteorological data. The BIOME-BGC model adds, besides VPD, also soil moisture to include the effect of water limitation on the canopy conductance and eventually on NPP. Mu et al. (2007) compared the BIOME-BGC model with MODIS and found that in dry regions, the VPD was insufficient to fully capture the water stress. Garbulsky et al. (2010) also concluded in his analysis of the variability of LUEs worldwide that vapour pressure deficit had a high dispersion in the relation with actual LUE and recommends the use of scalars like the actual/potential evapotranspiration or evaporative fraction to downregulate the LUE.

These scalars can be categorized in the "climatic supply/demand control on ecosystem productivity". The assumption is that potential ET represents the environmental demand for evapotranspiration. The actual ET is then the amount of water that is actually removed from the surface through evaporation from the soil and transpiration of the vegetation. If the vegetation suffers from any stress, the difference between actual ET and potential ET (PET) increases. The Penman-Monteith equation is often the basis for the estimation of potential evapotranspiration, using climate variables such as temperature and radiation. The estimation of actual ET is less common but can be for example derived from potential ET through water balance models using available soil moisture or by estimation of crop coefficients (e.g. Kc-factors of the FAO-AQUACROP model). The NPP model CASA (Carnegie-Ames-Stanford Approach) makes use of a water stress factor based on this relation of actual ET/potential ET to downscale the LUE. This approach covers particularly the above ground "climatic demand" which can be considered as the atmospheric water stress. To cover the entire picture, also the below ground supply of soil water to the plant should be taken into account. After all, even if there is a high above ground atmospheric water stress it does not necessary mean that the vegetation productivity is hampered, as long as there is a sufficient below ground supply of soil water to the plants roots.

The above methods typically focus on the atmospheric demand for water (vapour-pressure deficit), but do not account for the direct effects of low soil moisture. Stocker et al. (2017) has shown that soil moisture effects reduce global annual photosynthesis by ~15%, increase interannual variability by more than 100% across 25% of the global vegetated land surface, and amplify the impacts of extreme events on primary production. Soil-moisture balance models like the one in the REGCROP model of Gobin (2010) combine the above ground demand of water with the below ground availability. From these balances, drought stress indices can be derived. This promising approach – of combing above ground demand and below ground supply – originates from the field of regional crop modelling (e.g. REGCROP, AQUACROP, …). The main challenge is the upscaling of these soil water balance models to a generic global application.

Water availability can also be directly estimated from satellite data, e.g. by the combination of NDVI and surface temperature or by calculating water-sensitive vegetation index based on SWIR and NIR bands, e.g. Global Vegetation Moisture Index (GVMI, Ceccato et al., 2002).

The current CLMS global DMP originates from the original definition of Veroustrate et al. (1994) but was operationally improved within the MARSOP project. Parallel to this, already some attempts were made to improve the algorithm, especially by including a short-term water limitation. Verstraeten et al. (2006) proposed a water limitation in the C-Fix model by downscaling the LUE values through a stomatal regulating factor controlled by soil moisture availability and one controlled by a modified vapour pressure deficit (VPD). The required inputs are soil water availability (estimated through the Soil Water Index, SWI) for the first and evaporative fraction (EF, derived from latent and sensible heat fluxes) as equivalent for the latter. Also, minimum and maximum LUE values are required per biome and soil water properties per soil type. Despite reasonable results when compared to FLUXNET data, these developments were never operationally implemented at a global scale due to the relatively complexity and the requirement of several coefficients dependent on local environment and vegetation conditions.

Maselli et al. (2009) adopted the water limitation approach of the CASA model to downscale the LUE in the original DMP algorithm. This method makes use of the efficiency factor actual/potential evapotranspiration. Potential ET (PET) was computed from available meteorological data (radiation and temperature) by means of the empirical method of Jensen and Haise (1963) and for simplicity, actual ET was assumed to equal precipitation when it is lower than PET. The resulting DMP, calculated over forest sites, were confronted against flux measurements showing higher accuracies for this improved model compared to the original DMP, especially in summer months.

Recently, initial tests were done to include a water limitation in the DMP based on a soil water balance model of Gobin (2010). The model was run on ECMWF rainfall and potential evapotranspiration to derive a drought index for 23 cropland flux sites. The focus was on agricultural sites, as the DMP is mainly used for agronomic purposes. These initial tests showed that – at present – no sufficient evidence was found that such water stress factor undoubtedly enhances the current DMP for agricultural sites. The main reason is that a simplified version of the water balance model – needed to run on a global scale – didn't fully captured the complexity of irrigation management and other hydrological contributions in specific cropland sites. The relation between this drought index and its impact on the DMP needs further research before it can be implemented operationally. Other tests with water stress indices based on operationally available meteorological data such as ECMWF based Reference Evapotranspiration (ET0) and Meteosat Second Generation (MSG) based actual Evapotranspiration (ETa) didn't yielded satisfactory results. The inclusion of water stress in the DMP at global scale with operationally available data is not mature enough, hence it was excluded from the product. Without an explicit water limitation factor, the current DMP is expected to overestimate GPP during drought events and should be considered as a potential value rather than actual GPP.

3.3.5 Carbon Use Efficiency (CUE)

The next step is to estimate the ratio between NPP and GPP; that is the Carbon use efficiency (CUE), which indicates the efficiency of vegetation to convert absorbed CO_2 from the atmosphere to plant materials. The remaining part of the absorbed CO_2 is primarily released as autotrophic respiration (the metabolism by plants), and is closely related to phenology, amount of biomass, temperature and soil moisture (Atkin et al. 2015). It is now recognized that, plants invest also a fraction (which can be as much of 20%) of GPP into facilitating nutrient uptake and defense mechanisms. These carbon losses are typically overlooked in LUE models. Often autotrophic respiration is modelled following simple temperature dependence equations (Davidson, 2005). However, acclimation of autotrophic respiration to global warming suggest that the temperature dependence may vary over time, and that these simple equations may not be appropriate to describe autotrophic respiration (Davidson, et al., 2005; Ballantyne et al., 2017). An alternative is to use process based global dynamic vegetation models to estimate spatial and temporal variability in the CUE (Ardö et al., 2018), then to be applied to the modelled GPP for estimating biomass production. He et al. (2021) compared five process-based models with the MODIS CUE. The global average CUE derived from the five process-based models is 0.45 ± 0.05 (range from 0.38 to 0.52), slightly lower than the value of 0.48 obtained from MODIS data. A common strong latitudinal gradient of CUE was observed amongst the different datasets but considerable discrepancies in CUE estimations among where observed, especially in the temperate Northern Hemisphere. The authors highlight the need for a better understanding of the spatio-temporal dynamics of autotrophic respiration. To summarize we can say that the additional terms required to calculate autotrophic respiration as well as the additional carbon losses are, unfortunately, much less well understood from a theoretical and quantitative point of view than the terms in the equations for GPP itself. In the next two chapters the challenges of autotrophic respiration modelling are discussed in more detail.

Autotrophic respiration, a carbon dioxide loss from the plant, is the cost of producing metabolites that are used as building blocks for the synthesis of organic molecules. Even though approximately half of the gross photosynthetic assimilation is thought to be lost through autotrophic respiration, our current understanding of its drivers is limited. Moreover, little accurate reference data exist to calibrate and validate existing models. Hence, most observational and modelling studies emphasized the uptake process of CO_2 rather than the respiratory phase. Yet, various sub-models are developed for expressing plant respiration in vegetation models, from the non-explicit treatment of autotrophic respiration to attempts to develop more mechanistic respiration models (Gifford, 2003). The former is the most empirical. By not explicitly treating photosynthesis and respiration, parameters of vegetation models are directly calibrated against NPP measurements. Other models have a specific representation of autotrophic respiration, but the implementation can differ significantly. A classical approach is to use a constant GPP/Ra fraction, assuming only the environmental drivers of photosynthesis also determine the coupled respiration process. Autotrophic respiration models are

often divided in growth and maintenance respiration functions, where the first describes the energy cost needed for the accumulation of fresh biomass and the latter accounts for the basal mechanism to maintain the existing biomass (McCree, 1972; Penning De Vries, 1972). This theoretical framework is however criticized by Gifford (2003) who states that the distinction is only operational and no fundamental difference between the two can be made. Also Amthor (2000) prefers other theoretical frameworks but pleads for a more general paradigm to relate respiration to any number of individual processes.

The influence of temperature on respiration also remains an object of investigation. The concept of the Q10 temperature coefficient is widely used. Q10 is a measure of the growth rate as a consequence of increasing the temperature by 10° C, assuming the growth rate increases exponentially with temperature. The Q10 factor is often a constant near 2.0. Recent research pleads for a temperature adjustment of the Q10 coefficient (Tjoelker et al., 2001). Arrhenius-type models theoretically somewhat differ but in practice similar to the Q10 coefficient, are also used to implement the temperature sensitivity of autotrophic respiration (Kruse et al., 2011). Both approaches describe however only the short-term response of vegetation to the varying temperature. These models are often not explaining the observed variability in autotrophic respiration (Davidson, et al. 2006). Many researchers emphasized the importance of temperature acclimation of vegetation in addition to the instantaneous response of vegetation to changing temperatures (Smith and Dukes, 2013; Tjoelker et al., 2001; Wythers et al., 2005).

Given the challenges to model autotrophic respiration and to quantify the additional carbon losses, a more empirical approach is commonly suggested to estimate CUE. The CUE has been assumed to relatively constant across biomes and time, and a global-scale average is generally considered as 0.47 (Landsberg, et al., 2020; Waring et al., 1998). This assumption has been questioned; and for example, Delucia, et al. (2007) found that it varied between 0.23 and 0.83 for different forest depending on forest age. It has also been shown to vary depending on climatic variability (Chen & Yu, 2019). On the other hand, other studies based on a global data base show that CUE is independent of vegetation, environmental and climatic factors; and the main factor that causes the variability is the land management (Campioli, et al., 2015). Across natural ecosystems, the ratio of biomass production (closely related to NPP) to GPP was constant, suggesting that plant adaptation results in similar growth efficiency in high- and low-fertility natural systems. In managed lands, where there was a nutrient influx, a shift of the plants to a carbon investment in the aboveground biomass could be seen (Campioli et al., 2015). Vicca et al., (2012) also shows that soil fertility has the strongest impact on CUE. Landsberg et al. (2020) argue that models of biomass growth intended for use over large areas and long periods must be based on relatively simple relationships. It would be optimal if the stable CUE was based on appropriate physiological mechanisms than empirically established values; but we still lack a model including these physiological mechanisms. A constant or near constant CUE is currently an acceptable pragmatic alternative for global scale applications.

4 DMP ALGORITHM AND PRODUCT

4.1 HISTORY OF THE PRODUCT

A simplified Monteith approach, using remote sensing data to provide the fraction of absorbed PAR-radiation (fAPAR), was first implemented by Veroustraete (1994) and provided NPP estimates for European forests. This basic algorithm was consequently applied for entire Europe in the frame of the MARSOP project to produce DMP-estimates on a regular basis since around 2000, with unaltered parameterization. It was also implemented in the GLIMPSE software (Eerens et al., 2014). A parallel development, the C-Fix model – focusing on NPP – (Veroustraete et al., 2004, 2002; Verstraeten et al., 2006), tried to improve the basic algorithm with the implementation of a water stress factor, a temperature compartment model and biome specific LUE values. Despite reasonable success over Europe (Verstraeten et al., 2010), this model was never implemented in an operational context.

Over the years different versions of the DMP product arose which were distributed via separate channels to the user community. Since 2004, VITO produces global DMP-images derived from the 10-daily syntheses of SPOT-VEGETATION on behalf of the FOODSEC unit of JRC-MARS. Since 2007, part of this information is further distributed to users in Africa and South-America via the GeoNetCast service by another VITO-project called DevCoCast (or its predecessor VGT4Africa). In the summer of 2009, it was decided within the MARS-group to launch a new DMP version, using improved inputs (meteorological data and fAPAR). The MARSOP3 project ended in 2014 but the CLMS used the same climatology input data, algorithm and constants as for this latter MARSOP DMP. This version of the CLMS global DMP 1km is referred to as version 1 and is described in detail in GIOGL1_ATBD_DMP1km-V1.

Within CLMS, this original 1 km DMP (version 1) has evolved in a second version at 1 km, with following changes:

- Use of CLMS fAPAR 1km Version 2 as input
- Update of the CO_2 fertilization factor
- Fluxnet based biome specific LUEs
- Replacement of the autotrophic respiration factor

These developments were defined based upon suggestions in literature and the results from the validation of the 1km DMP version 1. This validation showed that the DMP version 1 was perturbed by cloud and snow contaminated values, expressed as flagged values – especially in winter in the northern hemisphere - or unreliable DMP values in regions with persistent cloud cover. In the 1 km fAPAR Version 2 data, such artefacts are eliminated by means of dedicated gap-filling procedures and a higher quality DMP is obtained. The use of biome specific light use efficiencies is based on the suggestion of several authors (e.g., Garbulsky et al., 2010; Turner et al., 2003) who claim that a biome-specific LUE is necessary in global DMP models. The validation of the DMP version 1 also

showed that biomes behaved differently when compared with reference data, e.g. cropland gave systematically lower GDMP values than the flux tower measurements and broadleaved evergreen forests were generally overestimated. This pattern already indicates that a biome-specific tuning of the LUE is appropriate. The same validation report also showed that the autotrophic respiration component in the DMP V1 produced unrealistic estimates. Hence it was chosen to replace this part of the model. More and more evidences are found in recent literature that the atmospheric CO_2 concentration impacts the global ecosystem productivity. So, this component of the model was also refined. The current version of the 300m DMP uses the same algorithm as the DMP 1 km version 2 [CGLOPS1_ATBD_DMP1km-V2].

The current product is treated as a potential or optimal value, representative for well-watered conditions. Previous efforts have shown that the accuracy of the DMP/NPP can somehow improve by adding a water stress factor (Maselli et al., 2009; Verstraeten et al., 2006). In the framework of CLMS, it was investigated whether adding a drought index based on a soil water balance could undoubtedly enhance the DMP of agricultural sites. Results from a study conducted for Australia by a research group using our DMP/NPP model showed promising results. But so far, initial tests aiming at global application of the same approach yielded not enough confidence to adopt this approach in the DMP and NPP.

4.2 INPUT DATA

4.2.1 Meteorological data

Global meteorological data are delivered by DTN providing for each “grid-cell” (at 0.25° resolution) the values of all standard meteorological variables. Basically, all daily data are “operational forecasts for the next 24 hours” derived from ECMWF HRES model).

The ECMWF climate data are used in different parts of the DMP algorithm:

- ***Radiation*** as basic input for the Monteith model.
- ***Temperature*** (daily minimum/maximum) in the temperature dependency for photosynthesis.

4.2.2 fAPAR

fAPAR corresponds to the fraction of photosynthetically active radiation absorbed by the green elements of the canopy. It depends on canopy structure, vegetation element optical properties and illumination conditions. The current 300 m DMP/NPP uses a 10-daily fAPAR based on PROBA-V data for products from January/2014 till June/2020 [CGLOPS1_ATBD_FAPAR300m-V2.0], and fAPAR based on Sentinel-3/OLCI from July/2020 onwards [CGLOPS1_ATBD_FAPAR300m-V2.0].

The algorithm, dedicated to the estimation of fAPAR from the PROBA-V and Sentinel-3/OLCI series of observations at 300m ground sampling distance, has been developed with the objective to provide high level of consistency with 1km products Version 2 derived from SPOT/VGT when aggregated at the 1 km resolution [CGLOPS1_ATBD_FAPAR1km-V2]. These LAI, fAPAR and FCOVER 300m products have the same temporal sampling frequency of 10 days. Products are associated with quality assessment flags as well as quantified uncertainties. The algorithm runs at the pixel level without interactions with the surrounding pixels.

There are two major changes according to the Version 2 algorithm:

- No SWIR bands are used since the SWIR band of PROBA-V has not the desired geometrical characteristics due to the resolution and ground spatial sampling distance which are roughly twice that of the VIS-NIR domain. Also, for the SWIR bands measured by Sentinel-3/SLSTR, the ground sampling distance is larger compared to Sentinel-3/OLCI. In addition, important calibration issues are reported for the SWIR bands 5 and 6 of SLSTR (Smith et al., 2019).
- It is not possible to use the climatology (or long-term averaged vegetation index profiles) as a background information since
    - The land cover at the 300m pixel scale may change significantly from one year to another for a significant fraction of the pixels.
    - The building of the climatology would require at least few years of input data, or, alternatively, should be derived from other observations including the MERIS or MODIS time series.

4.2.3 Land cover information

The DMP V2.0 and NPP V2.0 uses biome specific Light Use Efficiency (LUE) values. The information on the global distribution of land cover types comes from the ESA CCI Land Cover Map (epoch 2010) at 300m resolution, which was derived from ENVISAT-MERIS and SPOT-VGT imagery of the period 2008-2010. The land cover classes are based on the UN Land Cover Classification System (LCCS). The global map is shown in Figure 4.

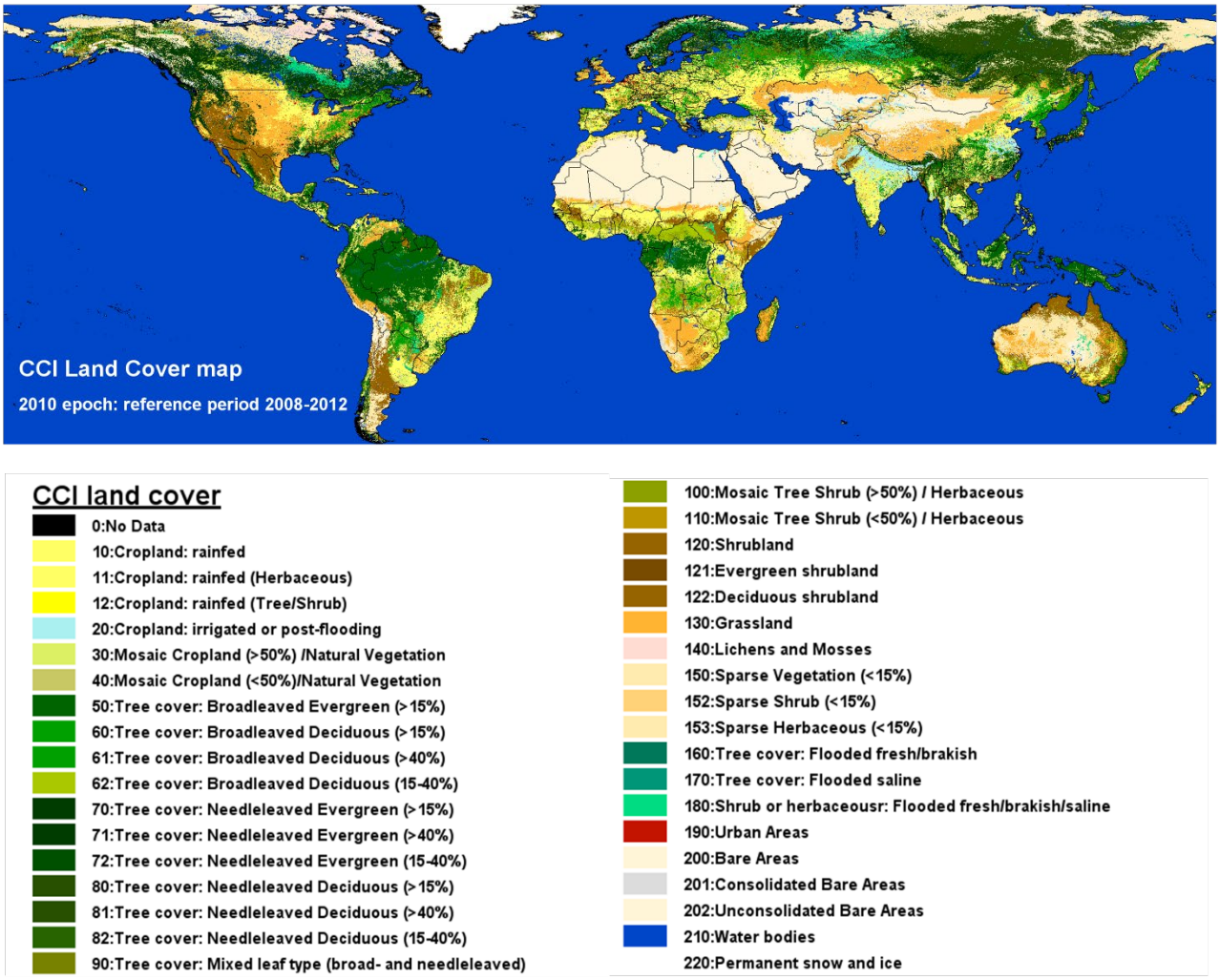


Figure 4: Global land cover map (ESA CCI epoch 2010) (top) and legend (bottom) derived from ENVISAT MERIS and SPOT-VGT data.

4.3 METHODOLOGY

4.3.1 Components of the GDMP/DMP

The GDMP and DMP product of the Global Land service is currently computed with the following Monteith variant (see Table 3):

$$GDMP = R \times \varepsilon_c \times fAPAR \times \varepsilon_{LUEc} \times \varepsilon_T \times \varepsilon_{CO2} [\times \varepsilon_{RES}] \quad (5)$$

$$DMP = R \times \varepsilon_c \times fAPAR \times \varepsilon_{LUEc} \times \varepsilon_T \times \varepsilon_{CO2} \times CUE[\times \varepsilon_{RES}] \quad (6)$$

NPP and GPP are derived from DMP and GDMP, respectively, by applying a conversion factor between dry matter and carbon and scaling according to the units.

$$\mathbf{1\ kgDM/ha/day = 1000\ gDM/ha/day = 0.1\ gDM/m^2/day} \quad \mathbf{(7)}$$

According to IPCC (IPCC, 2003), the efficiency of the conversion between carbon and dry matter is on the average 0.5 gC/gDM. So, NPP and DMP only differ by a constant fraction. In practice, to scale DMP to NPP the following calculation should be done:

$$\mathbf{NPP\ [gC/m^2/day] = DMP\ [kgDM/ha/day] \times 0.5 \times 0.1} \quad \mathbf{(8)}$$

The same relationships exist between GDMP and GPP.

Table 3: Individual terms in the Monteith variant used for Global Land service GDMP/DMP. All terms are expressed on a daily basis.

TERM	*MEANING*	*VALUE*	*UNIT*
GDMP	Gross Dry Matter Productivity	0 – 640	kgDM/ha/day
DMP	Dry Matter Productivity	0 – 320	kgDM/ha/day
R	Total shortwave incoming radiation (0.2 – 3.0µm)	0 – 320	GJ_T/ha/day
ε_c	Fraction of PAR (0.4 – 0.7µm) in total shortwave	0.48	J_P/J_T
*f*APAR	PAR-fraction absorbed (PA) by green vegetation	0.0 ... 1.0	J_{AP}/J_P
ε_{LUEc}	Light use efficiency (DM=Dry Matter) at optimum	Biome-specific	$kgDM/GJ_{AP}$
ε_T	Normalized temperature effect	0.0 ... 1.0	-
ε_{CO2}	Normalized CO_2 fertilization effect	0.0 ... 1.0	-
CUE	Carbon Use Efficiency	0.5	-
ε_{RES}	Fraction kept after omitted effects (drought, pests...)	1.0	-

These terms are explained separately in more detail below. Focus is made on the terms fAPAR, ε_{CO2} and ε_{LUEc}, which were modified in the version 2 DMP product.

- **Total shortwave incoming radiation**

The incoming solar (shortwave) radiation R is derived from ECMWF's global meteorological data from DTN at 0.25°, mostly reported in terms of $kJ_T/m^2/day$ with variations between 0 and 32,000, which corresponds to 320 $GJ_T/ha/day$ (1 hectare is 10 000m², and 1 GJ is 1 000 000 kJ).

- **The climatic efficiency**

The fraction ε_c of PAR (Photosynthetically Active Radiation, 0.4-0.7 µm) within the total shortwave (0.2 – 3.0µm) is set to a global constant of ε_c =0.48 (McCree, 1972).

- **Fraction of Absorbed Photosynthetic Active Radiation**

The *f*APAR is obtained from remote sensing observations (see also 4.2.2). These data are derived from PROBA-V or Sentinel-3/OLCI as 10-daily composites.

- **Biome specific LUEs**

In the first version of the GDMP at 1 km, a global LUE constant of 2.54 $kgDM/GJ_{AP}$ was applied. In this GDMP version 2 at 1 km, biome-specific LUE values are estimated by calibrating the GDMP with flux tower GPP measurements. These values are adopted in the 300m DMP and NPP product. Below briefly the procedure:

- The GDMPv2 at 1 km uses the CCI land cover information. For each flux site, the associated land cover class is derived from the global CCI map at 300 m.
- The FLUXNET data covers 224 towers with for each tower from 1 to more than 10 years of GPP measurements available and spread over different classes, see Figure 5.
- This FLUXNET dataset is per land cover divided into a calibration (75%) and validation (25%) dataset.
- For each flux site, the GDMPv2 is calculated with the towers ECMWF meteorological data and extracted fAPAR profiles from the global CLMS fAPAR Version 2 imagery, hereby excluding the LUE factor (LUE=1).
- Per land cover, an optimal LUE is found by fitting the GDMPv2 against the FLUXNET GDMP observations. A non-linear least squares fit was used to minimize the error between the flux tower GDMP and the CLMS GDMP. The resulting LUE values are shown in Table 4, which also includes the number of towers used per land cover, and number of observations used for the calibration and validation.
- For the land covers where no flux data were available, the LUE of similar classes are adopted. This occurs mainly for subclasses which adopt the LUE of their general class or for low-vegetated classes which adopt the LUE of the "Sparse Vegetation (<15%)" class. For the class "Urban Areas", an unreliable high LUE was found, probably due to a

misclassification of the flux site. Therefore, this class is also assigned as "Sparse Vegetation (<15%)". Theoretically, the LUE of "Permanent snow and ice" should be 0. However, due to misclassification in the land cover map, some pixels could still contain some vegetation as reflected by fAPAR. Then, it was chosen to assign the LUE of the "Sparse vegetation (<15%)".

- Figure 6 shows the global map when these LUE values are assigned to the CCI land covers.

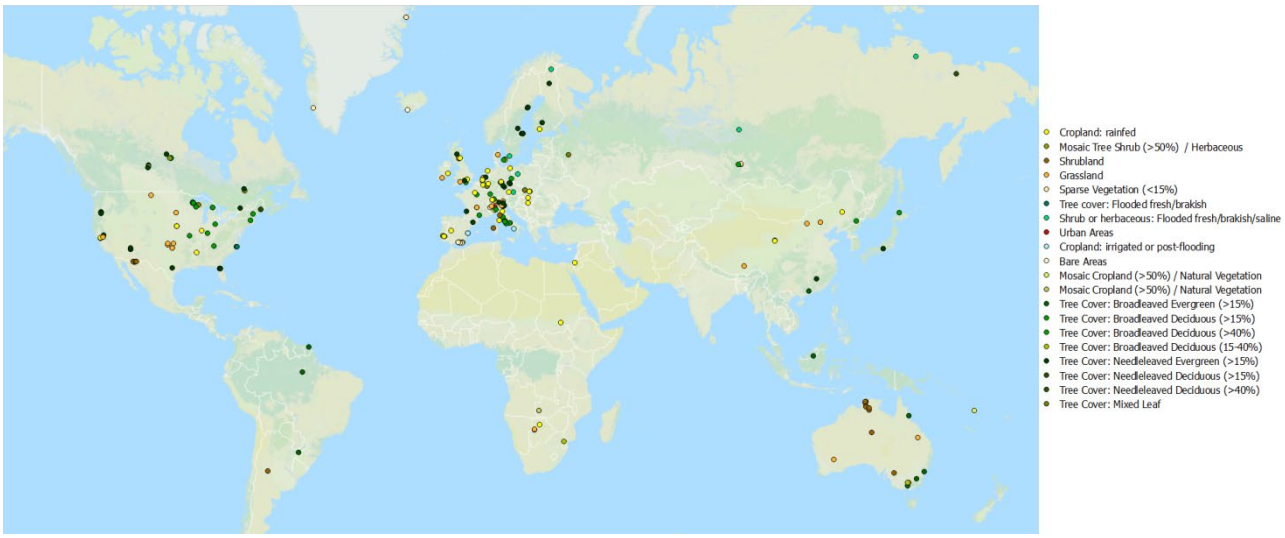


Figure 5: FLUXNET sites available for the calibration of the LUE parameter in the GDMPv2.

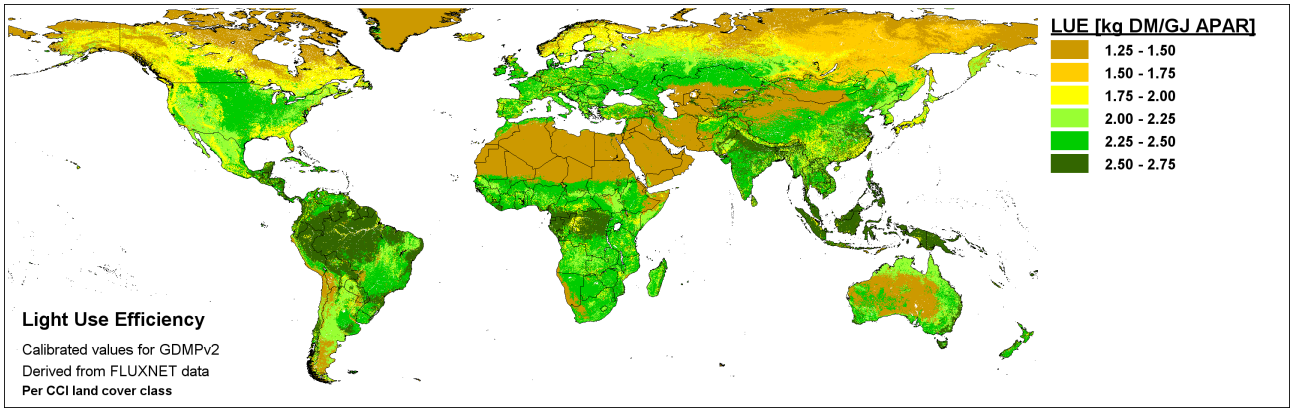


Figure 6: Global overview of the optimized Light Use Efficiency (LUE) values, calibrated with FLUXNET data, and assigned to the CCI land covers.

Table 4: ESA CCI land cover class with parameterized Light Use Efficiency (LUE) values. #towers are the number of towers available per land cover. #cal and #val are the number of observations used for calibration and validation. LUE is the optimized Light Use Efficiency by calibrating the 1km GDMPv2 against flux data. RMSEcal is the remaining RMSE after calibration. The colors of the table are linked to the LUE legend of Figure 6.

Lcid	LCabbr	LCname	#towers	#cal	#val	LUEopt	RMSEcal	Comment
0	NOD	NoData	0	0	0	1,42	n.a.	No flux-data. LUE of "SparseVegetation(<15%)"
10	CRO	Cropland-rainfed	45	4328	1443	2,49	74,39	
11	CRH	Cropland-rainfed-herbaceous	0	0	0	2,49	n.a.	No flux-data. LUE of "cropland-rainfed"
12	CRT	Cropland-rainfed-tree/shrub	0	0	0	2,49	n.a.	No flux-data. LUE of "cropland-rainfed"
20	CRI	Cropland-irrigated	2	165	56	2,70	72,14	
30	MCN	Mosaic-cropland(>50%)/naturalvegetation(tree	2	102	34	2,70	28,79	
40	MNC	Mosaic-naturalvegetation(>50%)	2	114	39	1,93	59,63	
50	BET	TreeCover-broadleaved-evergreen-closedtoopen(>15%)	8	747	249	2,56	60,19	
60	BDT	TreeCover-broadleaved-deciduous	26	2748	917	1,99	55,52	
61	BDC	TreeCover-broadleaved-deciduous-closed(>40%)	6	516	173	1,99	43,47	
62	BDO	TreeCover-broadleaved-deciduous-open(15-40%)	4	349	117	2,48	58,88	
70	NET	TreeCover-needleleaved-evergreen-closetoopen(>15%)	45	3827	1276	1,98	45,50	
71	NEC	TreeCover-needleleaved-evergreen-closed(>40%)	0	0	0	1,98	n.a.	No flux-data. LUE of "TreeCover-needleleaved-evergreen-closetoopen(>15%)"
72	NEC	TreeCover-needleleaved-evergreen-open(15-40%)	0	0	0	1,98	n.a.	No flux-data. LUE of "TreeCover-needleleaved-evergreen-closetoopen(>15%)"
80	NDT	TreeCover-needleleaved-deciduous-closetoopen(>15%)	1	18	6	1,57	16,03	
81	NDC	TreeCover-needleleaved-deciduous-close(>40%)	1	37	13	1,85	25,30	
82	NDC	TreeCover-needleleaved-deciduous-open(15-40%)	0	0	0	1,57	n.a.	No flux-data. LUE of "TreeCover-needleleaved-evergreen-closetoopen(>15%)"
90	MFT	TreeCover-mixedleaftype	6	893	298	2,23	42,88	
100	MTH	Mosaic-TreeShrub(>50%)-Herbaceous(<50%)	3	249	83	1,78	40,29	
110	MHT	Mosaic-Herbaceous(>50%)-TreeShrub(<50%)	0	0	0	1,78	n.a.	No flux-data. LUE of "Mosaic-TreeShrub(>50%)-Herbaceous(<50%)"
120	SHR	Shrubland	18	2010	670	2,10	44,52	
121	SHE	EvergreenShrubland	0	0	0	2,10	n.a.	No flux-data. LUE of "shrubland"
122	SHD	DecidousShrubland	0	0	0	2,10	n.a.	No flux-data. LUE of "shrubland"
130	GRA	Grassland	30	3158	1053	2,39	53,89	
140	LMO	LichensAndMosses	0	0	0	1,42	n.a.	No flux-data. LUE of "SparseVegetation(<15%)"
150	SPA	SparseVegetation(<15%)	10	588	197	1,42	37,40	
152	SPS	SparseShrub(>15%)	0	0	0	1,42	n.a.	No flux-data. LUE of "SparseVegetation(<15%)"
153	SPH	SparseHerbaceousCover(<15%)	0	0	0	1,42	n.a.	No flux-data. LUE of "SparseVegetation(<15%)"
160	TFF	TreeCover-flooded-freshorbrakishwater	3	226	76	1,87	53,08	
170	TFS	TreeCover-flooded-salinewater	0	0	0	1,87	n.a.	No flux-data. LUE of "TreeCover-flooded-freshorbrakishwater"
180	SFH	ShrubOrHerbaceous-flooded-fresh/saline/brakishwater	10	681	227	1,43	29,22	
190	URB	UrbanAreas	2	325	109	1,42	n.a.	Unreliable LUE. LUE of "SparseVegetation(<15%)"
200	BAR	BareaAreas	0	0	0	1,42	n.a.	No flux-data. LUE of "SparseVegetation(<15%)"
201	BAC	ConsolidatedBareAreas	0	0	0	1,42	n.a.	No flux-data. LUE of "SparseVegetation(<15%)"
202	BAU	UnconsolidatedBareAreas	0	0	0	1,42	n.a.	No flux-data. LUE of "SparseVegetation(<15%)"
210	WAT	Waterbodies	0	0	0	0,00	n.a.	No flux-data. LUE = 0
220	SNO	PermanentSnowAndIce	0	0	0	1,42	n.a.	No flux-data. LUE of "SparseVegetation(<15%)"

- **Normalized temperature effect**

The behaviour of ε_T is simulated with the model of Wang (1996) and is parameterized with data of Samson et al. (1997) for deciduous forests and of Jach and Ceulemans (2000) for a pine forest. The formula and parameters are listed below (Table 5):

$$p(T_d)\frac{e^{\left(c_1-\frac{\Delta H_{aP}}{R_g \times T_d}\right)}}{1+e^{\left(\frac{\Delta S \times T_d - \Delta H_{dP}}{R_g \times T_d}\right)}} \quad (9)$$

Table 5: List of the parameters used in the temperature function $p(T_d)$

TERM	*MEANING*	*VALUE*	UNIT
T_d	Air temperature (daily mean)	Variable	K
R_g	Universal gas constant	8.31	J/K/mol
c_1	Constant	21.77	-
ΔH_{aP}	Activation energy	52 750	J/mol
ΔH_{dP}	Deactivation energy	211 000	J/mol
ΔS	Entropy of the denaturation equilibrium of CO_2	704.98	J/K/mol

- **Normalized CO_2 fertilization effect**

The CO_2 fertilization effect or ε_{CO2} is defined as the increase in carbon assimilation due to CO_2 levels above the atmospheric background level (or reference level). This effect is influenced by the temperature and described by following formula:

$$F(CO_2, T_d) = \frac{[CO_2]-\frac{[O_2]}{2\tau}}{[CO_2]ref-\frac{[O_2]}{2\tau}} \times \frac{K_m \times \left(1+\frac{[O_2]}{K_0}\right)+[CO_2]ref}{K_m \times \left(1+\frac{[O_2]}{K_0}\right)+[CO_2]} \quad (10)$$

The temperature dependency shows two phases (Veroustraete, 1994) hence two sets of parameters are used in the model, depending on temperature conditions:

$$K_m = A_1 \times e^{(-E_{a1}/R_g \times T_d)} \quad \text{if } T_d \geq 288.13\text{K}$$

$$K_m = A_2 \times e^{(-E_{a2}/R_g \times T_d)} \quad \text{if } T_d \geq 288.13\text{K}$$

$$K_0 = A_0 \times e^{(-E_0/R_g \times T_d)} \quad (11)$$

$$\tau = A_\tau \times e^{(-E_\tau / R_g \times T_d)}$$

Table 6 describes the different parameters.

Table 6: List of the parameters used in the CO_2 fertilization factor.

TERM	*MEANING*	*VALUE*	UNIT
T_d	Air temperature (daily mean)	Variable	K
$[O_2]$	O_2-concentration	20.9	%
$[CO_2]_{ref}$	CO2-concentration in reference year 1832	281	ppmv
$[CO_2]$	actual CO2-concentration or "mixing ratio", eg. 355.61 (NB: ppmv = µmol/mol)	Variable	ppmv
R_g	Universal gas constant	8.31	J/K/mol
K_m	Affinity constant for CO_2 of Rubisco	Variable	$[\%CO_2]$
K_0	Inhibition constant for O_2	Variable	$[\%O_2]$
τ	CO_2/O_2 specificity ratio	Variable	[-]
E_{a1}	Constant in K_m-equation for T ≥ 288.13K	59 400	J/mol
A_1	Constant in K_m-equation	$2.419.10^{13}$	$[\%CO_2]$
E_{a2}	Constant in K_m-equation for T < 288.13K	109 600	J/mol
A_2	Constant in K_m-equation	$1.976.10^{22}$	$[\%CO_2]$
E_0	Constant in K_0-equation	13 913.5	J/mol
A_0	Constant in K_0-equation	8 240	$[\%O_2]$
E_τ	Constant in τ-equation	-42 869.9	J/mol
A_τ	Constant in τ-equation	$7.87.10^{-5}$	[-]

In the DMP and NPP 300m, we rely on recent studies (e.g. Zhu et al., 2016) which emphasize the role of atmospheric CO_2 fertilization in global ecosystem productivity. The atmospheric $[CO_2]$ concentration is implemented in the DMP 300m as a linear function of the calendar year. This function is based on a regression on data of annual 'spatial' average of globally averaged marine surface $[CO_2]$ data from the NOAA-ESRL cooperative air sampling network of the last 15 years. Figure 7 shows the measured and simulated global CO_2 concentrations, and the derived regression. This regression is alternatively expressed as:

$$[CO2] = 2.0775 \times YEAR - 3785.783 \quad (12)$$

For these last 15 years of global CO_2 data, the measurements and linearly simulated values agree very well. The [CO_2] data is obtained from ftp://aftp.cmdl.noaa.gov/products/trends/co2/co2_annmean_gl.txt.

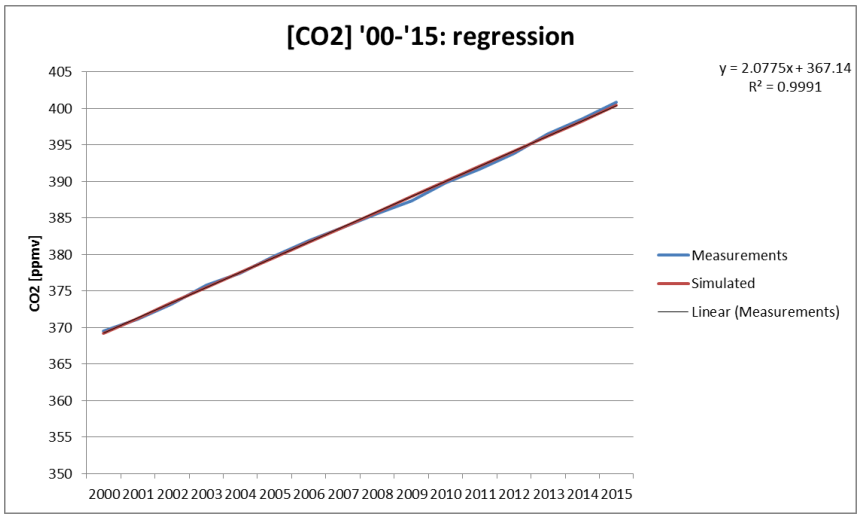


Figure 7: Global yearly atmospheric CO_2 measurements of the last 15 years, as measured by the NOAA-ESRL cooperative air sampling network and simulated with a yearly regression as used in the DMPv2

- **Carbon Use Efficiency**

The translation of Gross DMP in DMP (similar like GPP to NPP) can be summarized conceptually by the formula DMP (or NPP) = CUE x GDMP (or GPP) where CUE is the Carbon Use Efficiency. The additional terms required to calculate NPP from GPP are much less well understood from a theoretical and quantitative point of view than the terms in the equations for GPP itself. Attempts to model autotrophic respiration (e.g. such as in the DMPv1 at 1km, where autotrophic respiration is modelled as a simple linear function of daily mean air temperature Td, according to the parameterization of Goward and Dye (1987)) yield often unrealistic estimates and introduce significant uncertainties in the DMP/NPP products. There is not yet a mature global applicable methodology for estimating the GPP/NPP ratio. Therefore, it was decided to apply a constant CUE globally where DMP is considered to be half of the Gross DMP (CUE = 0.5).

- **Residuals**

The factor ε_{RES} (residual) is only added in equation (5) to emphasize the fact that some potentially important factors, such as drought stress, nutrient deficiencies, pests and plant diseases, are clearly

omitted here. As a consequence, the product might better be called “potential” DMP, because these stress factors are currently omitted. Especially for drought events, which has the strongest impact on GPP the current DMP is expected to overestimate biomass production. Only when the adverse effects of these residual vegetation stresses (drought, diseases and shortages of nutrients) impact the RS-derived fAPAR, it is captured by the DMP.

4.3.2 Practical procedure

4.3.2.1 Clamping fAPAR values

The input fAPAR product has small values over non vegetated areas, like deserts. This results in a positive DMP/GDMP/NPP/GPP value that are not realistic. Therefore, the low values of fAPAR are removed in the following way (see also Figure 8):

(1) All fAPAR values ≤0.05 are set to 0
(2) All fAPAR values >0.05 and <0.15 are rescaled linearly between 0 and 0.15
(3) All fAPAR values ≥0.15 remain unchanged.

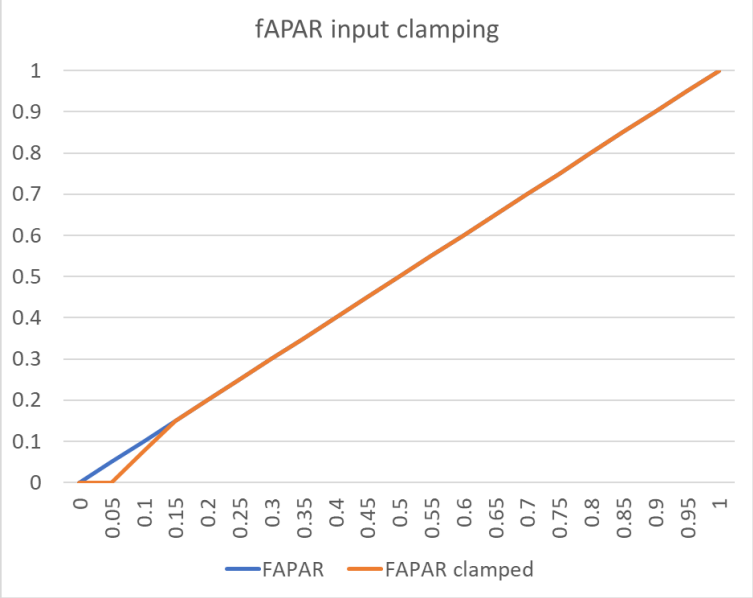


Figure 8: fAPAR clamping method.

4.3.2.2 Calculation of the GDMP/DMP

Given the simple elaboration of the epsilons (and dropping the factor ε_{RES}), equations (5) and (6) can be rewritten as:

$$\begin{aligned} \mathbf{GDMP} &= \mathbf{R \times \varepsilon c \times \mathit{f}APAR \times \varepsilon_{LUEc} \times \varepsilon_T \times \varepsilon_{CO2}} \\ &= \mathbf{\mathit{f}APAR \times \varepsilon_{LUEc} \times R \times \varepsilon(T,CO_2)} \\ &= \mathbf{\mathit{f}APAR \times \varepsilon_{LUEc} \times GDMP_{max}} \end{aligned} \quad \textbf{(13)}$$

$$\mathbf{DMP = GDMP \times CUE = GDMP \times 0.5} \quad \textbf{(14)}$$

with $\varepsilon(T,CO_2) = \varepsilon_c \times \varepsilon_T \times \varepsilon_{CO2}$ and ε_{LUEc} is the land cover specific LUE. This formulation better highlights the fact that (within the limits of the described model) GDMP is determined by four basic factors: *f*APAR, radiation (R), temperature (T) and CO_2. At the same time, equation (11) provides a practical method which allows bypassing the differences in temporal and spatial resolution between the inputs. In practice, the meteorological inputs (R, T) are provided on a daily basis and at a very low resolution (VLR), here 0.25° x 0.25°. On the contrary, the *f*APAR is provided on a dekadal basis, having pixels with a size of around 300m. Also, the final GDMP/DMP-product has this 300m resolution and dekadal frequency.

So, in practice we use the following procedure (modified from Eerens et al., 2004):

- Based on the meteorological inputs (R, T), the variable CO_2-level and the above-mentioned variant of the reduced (LUE=1) Monteith model (equations (5) and (11), equations (6) and (12)), images at 0.25° resolution are generated with

$$\mathbf{GDMP_{max} = R \times \varepsilon(T,CO2) = R \times \varepsilon_c \times \varepsilon_T \times \varepsilon_{CO2}} \quad \textbf{(15)}$$

- At the end of every dekad, we first compute a new image with the mean of the daily ($GDMP_{max,1}$) scenes. The resulting image ($GDMP_{max,10}$) is then resampled (bilinear interpolation) to the same resolution as the fAPAR image, derived from PROBA-V or Sentinel-3/OLCI (300m).
- Next, this $GDMP_{max}$ is multiplied with the LUE per land cover, and the 10-daily fAPAR, both at 300m resolution.
- The DMP is then calculated by multiplying the 10-daily GDMP with the autotrophic respiration coefficient, put to a constant fraction of 0.5 in this product version.

This practical approach (illustrated in Figure 9) can be formulated as follows (the subscripts 1 and 10 indicate daily and dekadal products, Nd is the number of days in each dekad)

$$\mathbf{GDMP_{10} = \mathit{f}APAR_{10} \times \varepsilon_{LUEc} \times GDMP_{max,10} \quad with \quad GDMP_{max,10} = \{\Sigma GDMP_{max,1}\} / N_d} \qquad \mathbf{(16)}$$

$$\mathbf{DMP_{10} = GDMP_{10} \times CUE \qquad with \quad CUE = 0.5} \qquad \mathbf{(17)}$$

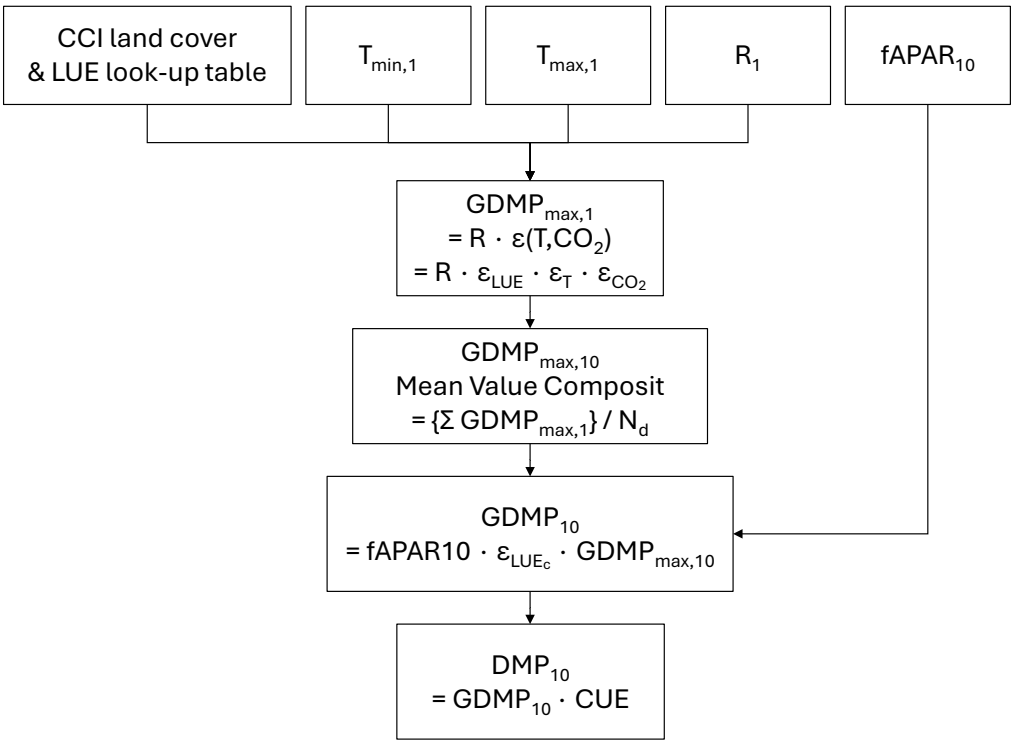


Figure 9: Process flow of GDMP/DMP. Based on meteorological data, a daily GDMPmax is estimated. At the end of each dekad, a Mean Value Composite of these $GDMP_{max}$ images is calculated. At the same time a fAPAR product is generated. The final $GDMP_{10}$ product is retrieved by the simple multiplication of the latter two images and, then, converted to DMP_{10}.

4.3.2.3 Calculation of GPP/NPP

GPP and NPP are derived from GDMP and DMP resp. using the following formulae:

$$\mathbf{GPP\ [gC/m^2/day] = GDMP\ [kgDM/ha/day] \times 0.5 \times 0.1} \qquad \mathbf{(18)}$$

$$\mathbf{NPP\ [gC/m^2/day] = DMP\ [kgDM/ha/day] \times 0.5 \times 0.1} \qquad \mathbf{(19)}$$

4.3.2.4 Near Real Time retrieval of the 300m DMP and NPP

The production of the DMP/GDMP and NPP/GPP is managed similarly to the production of fAPAR 300m. A brief description is given below but the reader is also referred to the product user manual of this fAPAR [CGLOPS1_PUM_FAPAR300m-V2.0]. The compositing for real time estimates is achieved according to the scheme described in Figure 10. We consider a particular dekadal date, *D*.

- In real time mode (RT0) (when the actual date corresponds to *D*, i.e. the first line in Figure 10), the compositing is achieved using only the past observations, with a maximum compositing window spanning c_- days in the past. The value of the product at dekadal date *D* is therefore relatively inaccurate.
- Then, the observations accumulate after dekadal date *D* when time passing. The value at dekadal date *D* is updated using the observations available after dekadal date *D* within the consolidation period. The accuracy of the product value progressively improves during this consolidation period.
- At the end of the consolidation period that lasts c_+ days, the value of the product value at dekadal date *D* have converged towards the 'historical' value, i.e. when there is at least c_- days (maximum length of the compositing window) after dekadal date *D*.

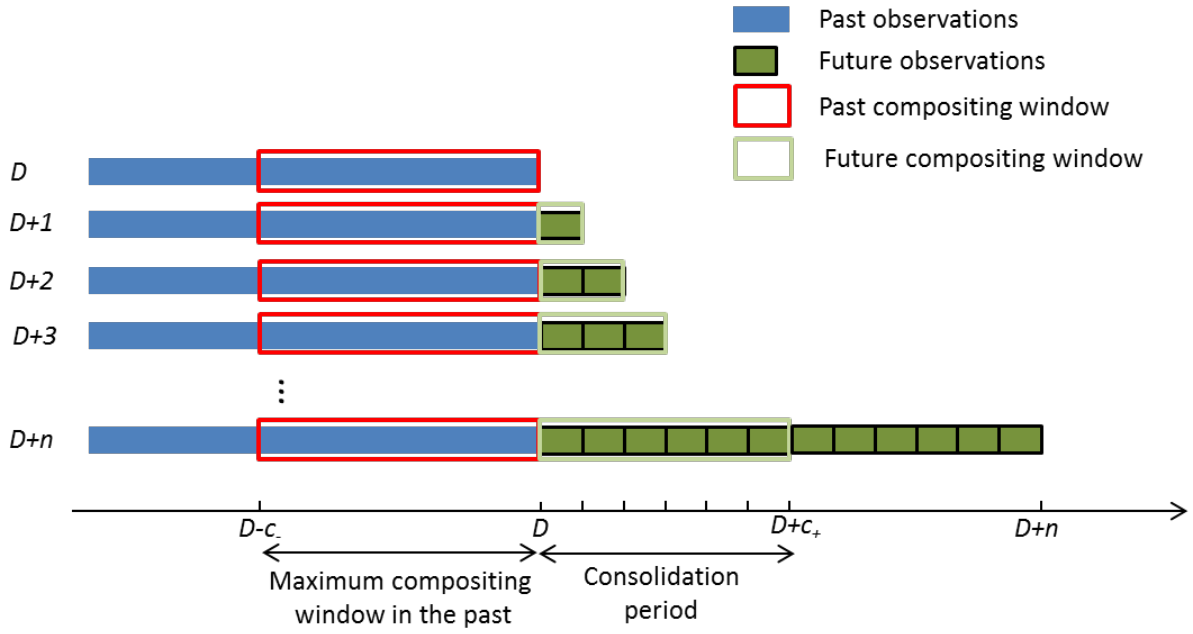


Figure 10: Scheme showing the compositing used for real time estimates

Note that the compositing period is not symmetric: the maximum compositing window in the past is longer than the maximum compositing window in the future (after date *D*) that is limited by the length of the consolidation period. The length of the maximum compositing window is set to 60 days, however more daily observations are used to classify the pixel as EBF (Evergreen Broadleaved Forests) or no-EBF. If not enough daily observations are available to estimate the decadal value, either an interpolation with the surrounding dekads is performed or in case of EBF the value is based on the previous decadal value.

Once a new dekad is available after dekadal date D, the value of the product at dekadal date D is updated and replaces the previous update. This step allows significant savings in storage. The reliability of the estimate increases with the consolidations. The product is provided in the following consolidations: RT0 (at dekad D), RT1 (at dekad D+1), RT2 (at dekad D+2) and RT6 (at dekad D+6).

4.4 OUTPUT PRODUCT

The algorithm produces dekadal global GDMP, DMP, GPP and NPP maps at 300 m (Figure 11). The physical DMP (GDMP) values range between 0 and 327.67 (655.34) kg dry matter per hectare per day and the physical NPP (GPP) values range between 0 and 16.3835 (32.767) kg carbon per square meter per day. The lowest values are found over non vegetated areas like deserts, urban areas or outside the growing season in cold climate zones. The highest values indicate highly productive pixels, often found over forested areas in the peak of the growing season or in tropical zones.

Some special values are used for missing data (-1) and for sea (-2).

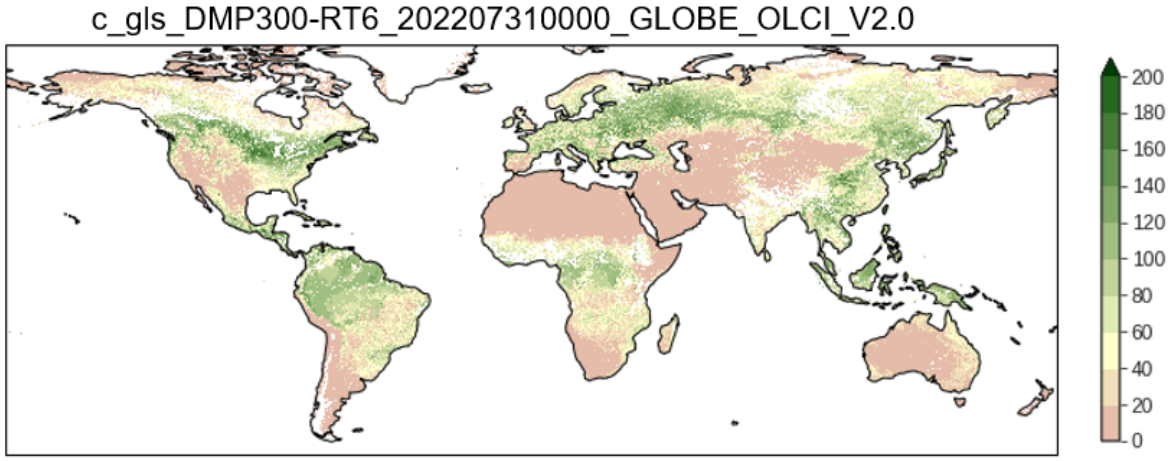


Figure 11: Global DMP at 300 m product for dekad 3 of July 2022. DMP is expressed in kg DM/ha/day.

The DMP, GDMP, NPP and GPP values are associated with a quality flag (Table 7) derived from the quality layer of the input fAPAR 300m product.

A full description of the product content and format can be found in the PUM [CGLOPS1_PUM_DMP-NPP300m-V2.0].

Table 7: Quality Flags of GDMP, DMP, GPP and NPP.

Bit	Qualitative indicator	Meaning	Mapping on fAPAR300
1	fAPAR missing or invalid	0 = Available fAPAR300 value 1 = Missing fAPAR300 value	fAPAR = 255
2	Meteorological data missing or invalid	0 = Available meteorological data values 1 = Missing meteorological data values	-
3	Not used		
4	Evergreen Broadleaf Forest (EBF)	1 = EBF 0 = no EBF	QFLAG bit 2
5	Not used		
6	Low FAPAR correction	1 = FAPAR<0.15 and rescaled according to Figure 8 0 = FAPAR is unchanged	
7	fAPAR computation for EBF	0 = fAPAR from daily observations 1 = fAPAR filled from previous dekad	QFLAG bit 5
8	fAPAR gap filling by interpolation	0 = fAPAR not interpolated 1 = fAPAR interpolated	QFLAG Bit 3

4.5 DIFFERENCES WITH THE PREVIOUS VERSION

The only difference between DMP 300m V1.0, DMP 300m V1.1, on one hand, and DMP 300m V2.0, on the other hand, is that V1.0 and V1.1 are derived from the fAPAR 300m V1.0, based upon PROBA-V data, and fAPAR 300m V1.1, based upon Sentinel-3 OLCI data, while V2.0 is derived from the fAPAR 300m V2.0 based upon PROBA-V data and the Sentinel-3 OLCI data. The

differences between fAPAR 300m V1.0, fAPAR 300m V1.1 and fAPAR 300m V2.0 are described in Section 4.5 of [CGLOPS1_ATBD_FAPAR300m-V2.0].

4.6 LIMITATIONS

Following remarks should be taken into consideration when interpreting the DMP (NPP) products:

- The biome-specific LUEs are calibrated against flux tower Eddy Covariance (EC) data. These flux data are considered as ground truth measurements of GPP. But in fact, this GPP is a combination of Net Ecosystem Exchange (NEE) measurements and a modelled respiration factor. Additionally, there are other reasons to question the relation between flux measurements and modelled GPP estimates. The spatial representativeness of the tower, for example, is strongly dependent on local conditions like the towers footprint, wind direction, etc. Thus, caution should be made when comparing 1 km pixel values with tower observations. Each tower also has its specific environmental conditions, like soil properties, precipitation, elevation, …This makes it difficult to generalize tower specific parameterization into broad categories like land cover.
- The biome-specific parametrization requires accurate information on the land cover type. At global scale, a land cover map always contains a fraction of falsely classified pixels.
- The in-situ data availability can determine the reliability of the biome-specific LUEs. For example, for needle leaved deciduous forests, there is only one tower available which even was classified differently in the land cover map as the IGBP legend.
- Some potentially important factors, such as drought stress, nutrient deficiencies, pests and plant diseases, are omitted in the DMP (NPP) product. As a consequence, the product might better be called “potential” DMP or NPP. On the other hand, it might be argued that the adverse effects of drought, diseases and shortages of nutrients are manifested (sooner or later) via the RS-derived fAPAR.
- The DMP algorithm does not include a water stress factor to account for short term drought stresses. In drought sensitive vegetation types, this can lead to an overestimation of the actual plant productivity. The temperature dependency factor uses generalized values derived from the parameterization on European forests (Veroustraete et al., 2002) and these parameterization may not hold for other biome types and agro-ecological zones. Especially the difference between C3 and C4 vegetation types is pronounced. C3 plants (like the forests used in the current calibration) have their optimal temperate around 20-25 degrees whereas C4 plants reach their optimum around 30-35 degrees (Massad et al., 2007; Yamori et al., 2014).
- fAPAR is included in Light Use Efficiency (LUE) models such as the DMP (NPP) to represent in a relative way an observation of the actual uptake of solar radiation by plants. Combined with the modelled response of vegetation LUE to environmental conditions (represented by the meteorological data as model input) satellite based GDMP (GPP) is estimated. The assumed

relation between fAPAR and GDMP (GPP) fits fairly well in general but there are a number of known cases where satellite based fAPAR is not capturing sufficiently accurate GPP or GDMP dynamics:

- Cloud contamination can result in changes in VIs that can confound the real seasonality of the vegetation.
- Optical sensors lack the capability to detect the full canopy activity of dense vegetation such as tropical forests.
- Vegetation greenness as reflected by fAPAR is not able to capture seasonal photosynthetic activity of evergreen trees because of the decoupling of greenness and photosynthesis (Huang et al., 2019).
- A similar decoupling can occur over wintergreen vegetation such as cover crops where fAPAR values are high but there is little or none photosynthetic activity and plant growth. This can result in overestimations of GDMP/DMP or GPP/NPP.
- In general, the spectral response of vegetation canopy is insufficient to capture drought effects of GDMP or GPP. Furthermore, the spectral sensitivity of vegetation canopies differs between ecosystems. Trees which have deeper roots are most resistant to decreased soil water and have a lower sensitivity, while the sensitivity is generally higher for cropland, grassland and shrubland (Zhang et al., 2016).

- The autotrophic respiration is calculated as a simple fraction of GDMP or GPP and is therefore assumed to have the same ecophysiological behaviour. In other models (e.g. used for MODIS product retrieval), the autotrophic respiration is considered as an independent component.
- The CO_2 fertilization factor relies on the original definition by Veroustraete (1994). We acknowledge that more sophisticated and state-of-the-art CO_2 fertilization submodels should be developed. But in the Copernicus Land Monitoring Service, the focus is on algorithm implementation rather than on fundamental research on algorithm components. Hence, we have to rely on existing described methods. If state-of-the-art modules are being developed that perform reliably better than the current approach, then it can be considered to implement them in a future version of the GDMP and GPP algorithm.

4.7 QUALITY ASSESSMENT

No standard procedure exists to assess the performance of EO-derived DMP products. Therefore, a procedure adapted from the guidelines, protocols and metrics defined by the Land Product Validation (LPV) group of the Committee on Earth Observation Satellite (CEOS) for the validation of satellite-derived land products is used. This processing includes checking the spatial consistency, the temporal consistency, performing a global statistical analysis and assessing the accuracy of the product.

The details of the procedure and the results are gathered into the Validation report [CGLOPS1_VR_DMP-NPP300m_V2.0].

4.8 RISK OF FAILURE AND MITIGATION MEASURES

With the present model for DMP estimation, there are the following risks:

- If no Sentinel-3/OLCI data are available, the DMP and NPP 300m cannot be generated. An alternative data source is the METOP-AVHRR but this only delivers fAPAR values at 1 km resolution.
- The DMP chain uses meteorological data based on the daily runs of the ECMWF High Resolution model and delivered in a specified format by a third party. We have now contractual assurance that any missing data can be delivered within 3 working days, given that the ECMWF data is available of course. Gaps of one or two days do not constitute a blocking issue as they are middled out over a dekad having a negligible effect on the end product. In the event of bigger gaps and the non-compliance of our contractor, we could still develop an alternative in house based on the ECMWF raw data. This will then mean a delay of a few weeks.

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